

A protein, as we have seen, is a polypeptide chain folded into one or more domains, each of which is made up of α helices, β sheets and loops. The process by which a polypeptide chain acquires its correct three-dimensional structure to achieve the biologically active native state is called protein folding. Although some polypeptide chains spontaneously fold into the native state, others require the assistance of enzymes, for example, to catalyze the formation and exchange of disulfide bonds; and many require the assistance of a class of proteins called chaperones. A chaperone binds to a partly folded polypeptide chain and prevents it from making illicit associations with other folded or partly folded proteins, hence the name chaperone. A chaperone also promotes the folding of the polypeptide chain it holds. After a polypeptide has acquired most of its correct secondary structure, with the α -helices and β -sheets formed, it has a looser tertiary structure than the native state and is said to be in the molten globular state. The compaction that is necessary to go from the molten globular state to the final native state occurs spontaneously.

Protein folding generates a particular three-dimensional structure from an essentially linear, one-dimensional structure—a polypeptide chain with a particular sequence of amino acid residues. How to predict the three-dimensional structure of a protein from its amino acid sequence is the major unsolved problem in structural molecular biology. If we had a general solution to the protein folding problem, it would be possible to write a computer program to simulate protein folding and generate the precise three-dimensional structure of any protein from its amino acid sequence. However, a general solution to the folding problem is still not in sight, even though the number of proteins whose three-dimensional structure has been solved experimentally, in other words, the database of known protein structures, is doubling every 2 years.

A protein in its native state is not static. The secondary structural elements of the domains as well as the entire domains continually undergo small movements in space, either fluctuations of individual atoms or collective motions of groups of atoms. Furthermore, the functional activities of many proteins depend upon large conformational changes triggered by ligand binding. In this chapter, after discussing protein folding, we shall examine some examples of functionally important conformational changes of proteins.

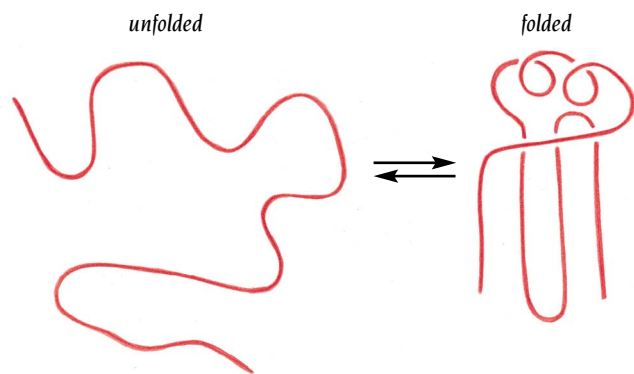


Figure 6.1 A polypeptide chain is extended and flexible in the unfolded, denatured state whereas it is globular and compact in the folded, native state.

Globular proteins are only marginally stable

Every biochemist or molecular biologist who has worked with proteins knows by experience that they are unstable. Slight changes in pH or temperature can convert a solution of biologically active protein molecules in their **native state** to a biologically inactive **denatured state**. The energy difference between these two states in physiological conditions is quite small, about 5–15 kcal/mol, not much more than the energy contribution of a single hydrogen bond, which is of the order of 2–5 kcal/mol.

There are two major contributors to the energy difference between the folded and the denatured state: enthalpy and entropy. **Enthalpy** derives from the energy of the noncovalent interactions within the polypeptide chain—the hydrophobic interactions, hydrogen bonds and ionic bonds. The covalent bonds within and between the amino acid residues in the polypeptide chain are the same in the native and denatured states, with the exceptions of disulfide bonds in those proteins where these form between cysteine residues. The noncovalent interactions on the other hand differ significantly between the two states. In the native state these interactions are maximized to produce a compact globular molecule with a tightly packed hydrophobic core whereas the denatured state is more open and the side chains are more loosely packed (Figure 6.1). These noncovalent interactions are therefore stronger and more frequent in the native state and hence their energy contribution, enthalpy, is much larger. The enthalpy difference between native and denatured states can reach several hundred kcal/mol.

Entropy derives from the second law of thermodynamics which states that energy is required to create order. Proteins in the native state are highly ordered in one main conformation whereas the denatured state is highly disordered, with the protein molecules in many different conformations. A typical experimental preparation of unfolded protein (a solution in 6 M guanidinium chloride or 8 M urea) contains 10^{15} – 10^{20} protein molecules, each of which will have a unique conformation. In the absence of compensating factors it would therefore be entropically much more favorable for the protein to be in the disordered denatured state. The energy difference due to entropy between the native ordered state and the denatured state can also reach several hundred kcal/mole but in the opposite direction to the enthalpy difference. The total energy difference between the native and the denatured state of 5–15 kcal/mol, which is called the **free energy** difference, is thus a difference between two large numbers, the enthalpy difference and the entropy difference. The fact that this difference is very small is a severe complicating factor both for predictions of possible native states and for interpretation of factors responsible for the stability or instability of protein molecules, because our knowledge about the denatured state is very incomplete.

We know much more about factors that influence the stability of the native state, mainly from experiments using directed mutations in proteins of known three-dimensional structure. Such experiments have yielded

precise information about energy contributions to the stability of the native state from close packing of hydrophobic side chains in the interior of the protein, and from the presence of disulfide bridges and interior hydrogen bonds and salt bridges, as well as from side chains that compensate the dipole moment of α helices (see Chapter 17).

The marginal stability of the native state over the denatured state is biologically very important. Living cells need globular proteins in correct quantities at appropriate times. It is therefore as important to be able easily to degrade these proteins as it is to be able to synthesize them. Globular proteins in living cells usually have a rather rapid turnover and their native states have therefore evolved to be only marginally stable. Moreover, the catalytic activities of enzymes, and other important functions of proteins, generally require some structural flexibility, which would be inconsistent with a rigidly stabilized structure.

Kinetic factors are important for folding

High resolution x-ray structure determinations of several hundred proteins have shown that in each case the specific sequence of a polypeptide chain appears to yield only a single, compact, biologically active fold in the native state. This fold generally has many substates with minor structural differences between them, as will be discussed later in this chapter, but all of these substates have the same general fold. Comparisons with structure determinations in solution by NMR show that the same fold also prevails in solution. In other words, under physiological conditions there appears to be one conformation for a given amino acid sequence that has a significantly lower free energy than any other. How is this folded state reached?

Intuitively one might imagine that all protein molecules search through all possible conformations in a random fashion until they are frozen at the lowest energy in the conformation of native state. The biophysicist Cyrus Levinthal showed in 1968 by a simple calculation that this is impossible. Assume as a gross simplification that each peptide group has only three possible conformations, the allowed regions α , β and L in the Ramachandran diagram (see Figure 1.7), and that it converts one conformation into another in the shortest possible time, one picosecond (10^{-12} seconds). A polypeptide chain of 150 residues would then have $3^{150} = 10^{68}$ possible conformations. To search all these conformations would require 10^{48} years (10^{56} seconds)—an astronomical number compared with the actual folding time, which is between 0.1 and 1000 seconds both *in vivo* and *in vitro*. To occur on this short time scale, the folding process must be directed in some way through a kinetic pathway of unstable intermediates to escape sampling a large number of irrelevant conformations.

Such a folding mechanism raises several important questions that are difficult to examine experimentally, since the possible intermediates have a very short lifetime. If kinetic factors are important for the folding process it is possible that the observed folded conformation is not the one with the lowest free energy but rather the most stable of those conformations that are kinetically accessible. The protein might be kinetically trapped in a local low energy state with a high energy barrier that prevents it from reaching the global energy minimum which might have a different fold. In such a case structure prediction by energy calculations would give the wrong structure even if such calculations could be made with great accuracy. One important question therefore is how a living cell can prevent the folding pathway from becoming blocked at an intermediate stage. The most common obstacles to correct folding seem to be (1) aggregation of the intermediates through exposed hydrophobic groups, (2) formation of incorrect disulfide bonds, and (3) isomerization of proline residues. To circumvent these three obstacles cells produce special proteins that assist the folding process, as we shall discuss later in this chapter.

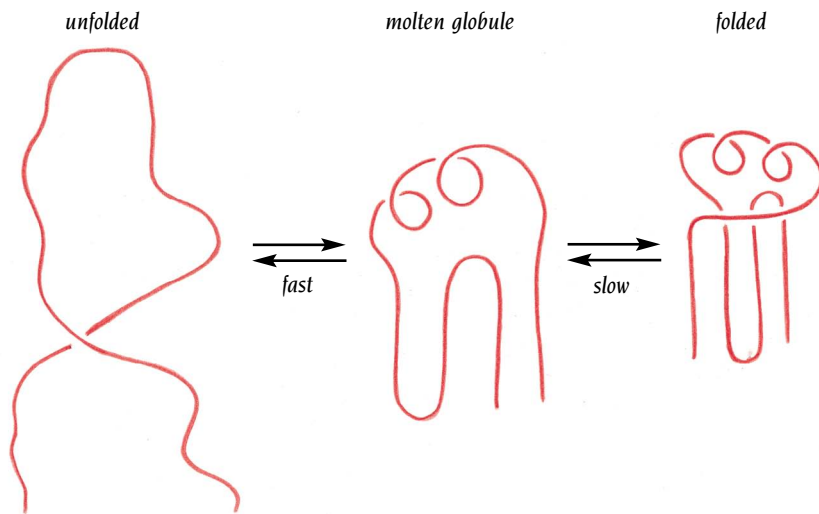


Figure 6.2 The molten globule state is an important intermediate in the folding pathway when a polypeptide chain converts from an unfolded to a folded state. The molten globule has most of the secondary structure of the native state but it is less compact and the proper packing interactions in the interior of the protein have not been formed.

An alternative way to remove kinetic barriers is exemplified by α -lytic protease, a bacterial enzyme which belongs to the serine protease superfamily of enzymes (Chapter 11). Like many other proteases it is synthesized and folded *in vivo* as an inactive precursor protein with a prosegment of 77 residues. This segment is excised after folding to produce the active enzyme. Unfolded precursor protein refolds easily *in vitro* but unfolded α -lytic protease lacking the prosegment does not refold. However, a solution of unfolded enzyme can be induced to refold by adding the excised prosegment. The capacity for folding obviously exists in the unfolded enzyme but there is a barrier present somewhere in the folding pathway that prevents folding. The prosegment removes this kinetic barrier, presumably by interacting with the enzyme in the unfolded state and thereby lowering the free energy of the transition states for folding; just as enzymes lower the free energy of transition states for chemical reactions and thereby increase the rates of the reactions (see Chapter 11).

Molten globules are intermediates in folding

The first observable event in the folding pathway of at least some proteins is a collapse of the flexible disordered unfolded polypeptide chain into a partly organized globular state, which is called the **molten globule** (Figure 6.2). This event is fast, usually within the deadtime of the experimental observation, which is a few milliseconds. We therefore know almost nothing about the process that leads to the molten globule, but we know some of the properties of this state. The molten globule has most of the secondary structure of the native state and in some cases even native-like positions of the α helices and β strands. It is less compact than the native structure and the proper packing interactions in the interior of the protein have not been formed. The interior side chains may be mobile, more closely resembling a liquid than the solid-like interior of the native state. Also loops and other elements of surface structure remain largely unfolded, with different conformations. The molten globule should, therefore, not be viewed as a single structural entity but as an ensemble of related structures that are rapidly interconverting (see Figure 6.3a).

In a second step, which can last up to 1 second, persistent native-like elements of tertiary structure begin to develop, possibly in the form of subdomains that are not yet properly docked. The ensemble of conformations is much reduced compared with those of the molten globule but it is still far from a single form. The single native form is reached in the final stage of folding, which involves the formation of native interactions throughout the protein, including hydrophobic packing in the interior as well as the fixation of surface loops.

Burying hydrophobic side chains is a key event

The collapse of the unfolded state to generate the molten globule embodies the main mystery of protein folding. What is the driving force behind the choice of native tertiary fold from a randomly oriented polypeptide chain?

There is very little change in free energy by forming the internal hydrogen bonds that are characteristic of α helices and β sheets because in the unfolded state equally stable hydrogen bonds can be formed to water molecules. Secondary structure formation therefore cannot be the thermodynamic driving force of protein folding. On the other hand there is a large free energy change by bringing hydrophobic side chains out of contact with water and into contact with each other in the interior of a globular entity. Thus the most likely scenario is that the polypeptide chain begins to form a compact shape with hydrophobic side chains at least partially buried very early in the folding process. This scenario has several important consequences. It vastly reduces the number of possible conformations that need to be searched because only those that are sterically accessible within this shape can be sampled. Second, when some of the side chains are partly buried, their polar backbone -NH and -CO groups are also buried in a hydrophobic environment unable to form hydrogen bonds to water. This is energetically unfavorable unless they form hydrogen bonds to each other, which they can only do if they are close together. The simplest way to form such bonds is by forming elements of secondary structure: α helices and β sheets. The formation of secondary structure early in the folding process can therefore be regarded as a consequence of burying hydrophobic side chains and not as a driving force for the formation of the molten globule.

Looking at the amino acid sequence of a globular protein one finds that hydrophobic side chains are usually scattered along the entire sequence in a seemingly random manner. In the native state of the folded protein about half of these side chains are buried in the interior and the rest are scattered on the surface of the protein, surrounded by hydrophilic side chains. The buried hydrophobic side chains are not clustered in the sequence but are scattered along the entire polypeptide chain. What causes these residues to be selectively buried during the early and rapid formation of the molten globule? This question must be answered before one can solve the folding problem and be able to predict the fold of a protein from its amino acid sequence.

Both single and multiple folding pathways have been observed

In order to understand fully any folding pathway, all states of the pathway must be characterized both structurally and energetically. The simplified diagram in Figure 6.3 illustrates that during the folding process the protein proceeds from a high energy unfolded state to a low energy native state through metastable intermediate states with local low energy minima separated by unstable transition states of higher energy. The characterization of these states is not trivial and many different experimental techniques are employed, including NMR, hydrogen exchange, spectroscopy and thermochemistry.

Recently Alan Fersht, Cambridge University, has developed a protein engineering procedure for such studies. The technique is based on investigation of the effects on the energetics of folding of single-site mutations in a protein of known structure. For example, if minimal mutations such as Ala to Gly in the solvent-exposed face of an α helix, destabilize both an intermediate state and the native state, as well as the transition state between them, it is likely that the helix is already fully formed in the intermediate state. If on the other hand the mutations destabilize the native state but do not affect the energy of the intermediate or transition states at all, it is likely that the helix is not formed until after the transition state.

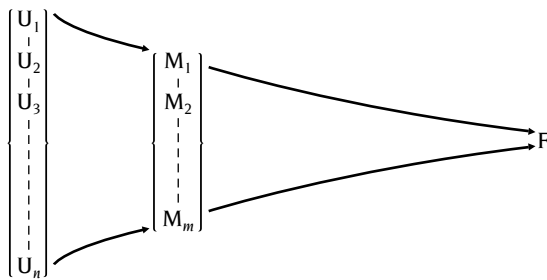
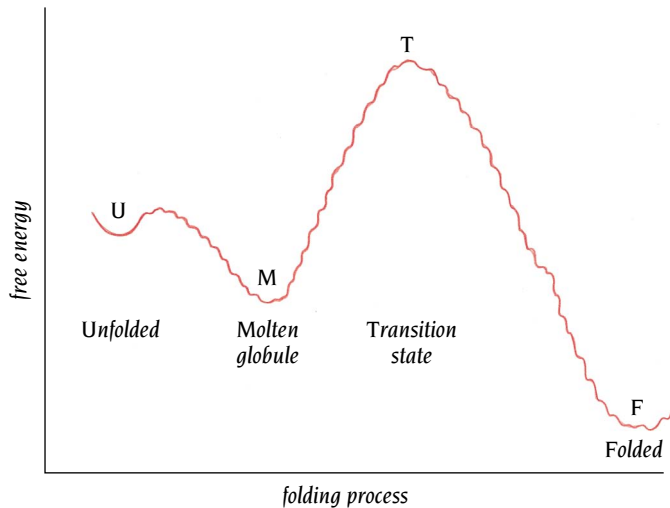


Figure 6.3 The unfolded state is an ensemble of a large number of conformationally different molecules, $U_1...U_n$, which undergo rapid interconversions. The molten globule is an ensemble of structurally related molecules, $M_1...M_m$, which are rapidly interconverting and which slowly change to a single unique conformation, the folded state F. During the folding process the protein proceeds from a high energy unfolded state to a low energy native state. The conversion from the molten globule state to the folded state is slow and passes through a high energy transition state, T.



The small bacterial ribonuclease, **barnase**, is a single chain protein with 110 amino acids and no disulfide bridges. Its three-dimensional structure was determined by the group of Guy Dodson, York University, and comprises three amino terminal α helices and a carboxy terminal five-stranded antiparallel β sheet (Figure 6.4). The group of Alan Fersht have examined the effects of mutations all along the structure and have made a detailed residue by residue characterization of its folding intermediate and transition states. They have concluded from their results that the intermediate molten globule state already has not only most of the native secondary structure elements but also the native-like relative positions of the α helix and β sheet as well as

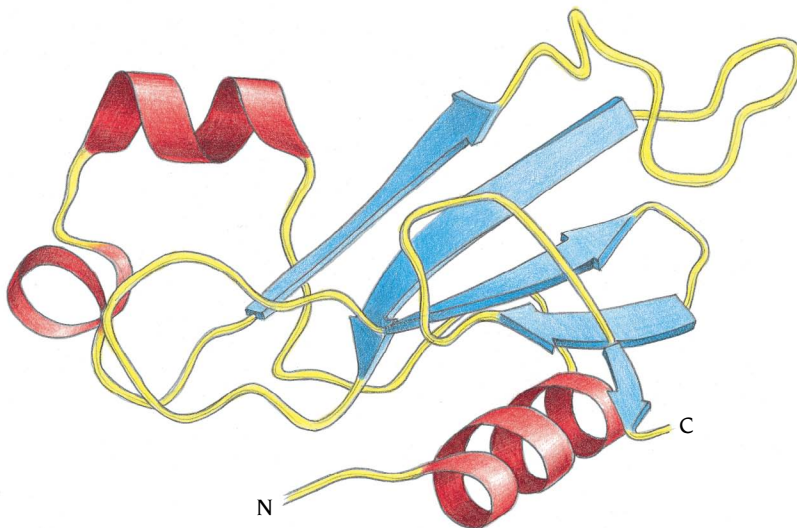


Figure 6.4 Schematic diagram of the structure of the enzyme barnase which is folded into a five stranded antiparallel β sheet (blue) and two α helices (red).

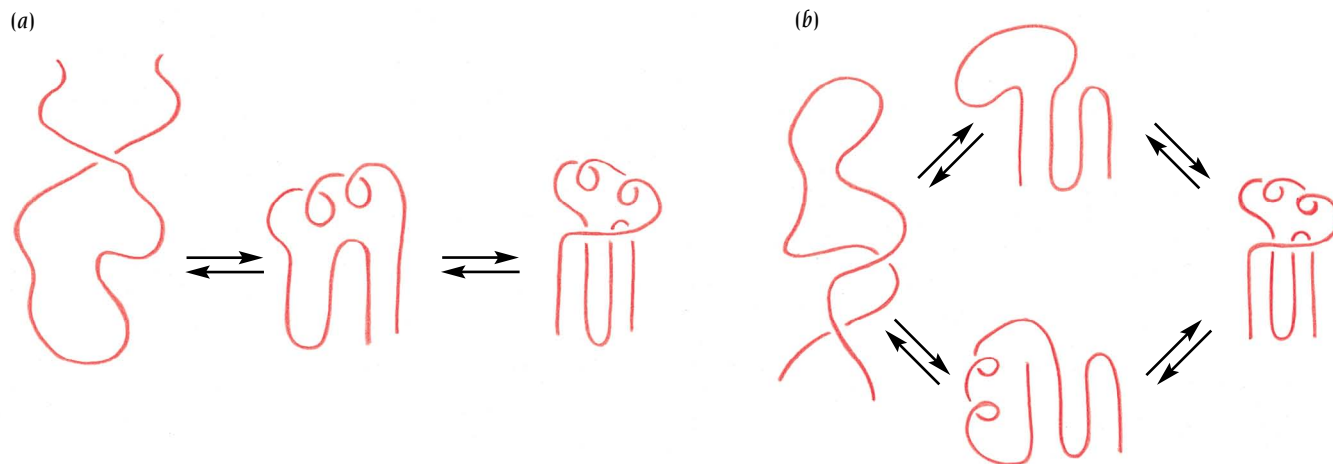


Figure 6.5 (a) Some proteins such as barnase fold through one major pathway whereas others fold through multiple pathways. (b) The folding of the enzyme lysozyme proceeds through at least two different pathways.

the relative positions of the β strands within the sheet. These results are consistent with the notion that the folding of barnase proceeds through a single major transition state and consequently through one major pathway (Figure 6.5a).

In contrast, folding of the enzyme **lysozyme** involves parallel pathways and distinct folding domains. Hen egg-white lysozyme was the first enzyme to have its structure determined crystallographically, in the laboratory of David Phillips then at the Royal Institution, London in 1965. The native structure consists of two lobes separated by a cleft (Figure 6.6). The first lobe comprises five α helices and the second is predominantly a three-stranded antiparallel β sheet. The folding of lysozyme has been studied extensively by a variety of complementary techniques (NMR, circular dichroism, fluorescence, hydrogen–deuterium exchange) to follow the development of different aspects of the structure such as formation of secondary structure, burial of hydrophobic aromatic groups and formation of hydrogen bonds. The group of Christopher Dobson, Oxford University, has used pulsed amide hydrogen–deuterium exchange to follow secondary structure formation. Amide hydrogen atoms are readily exchanged with the solvent in unfolded proteins, but this exchange is often strongly inhibited in a folded protein, especially for those amide groups that are hydrogen bonded in secondary structure elements. As a result, by measuring the rate of amide–hydrogen exchange as a function of folding time it is possible to monitor the formation of structure during the folding reaction. At 20 milliseconds, two major intermediate stages of lysozyme were detected: one in which the α -helical domain

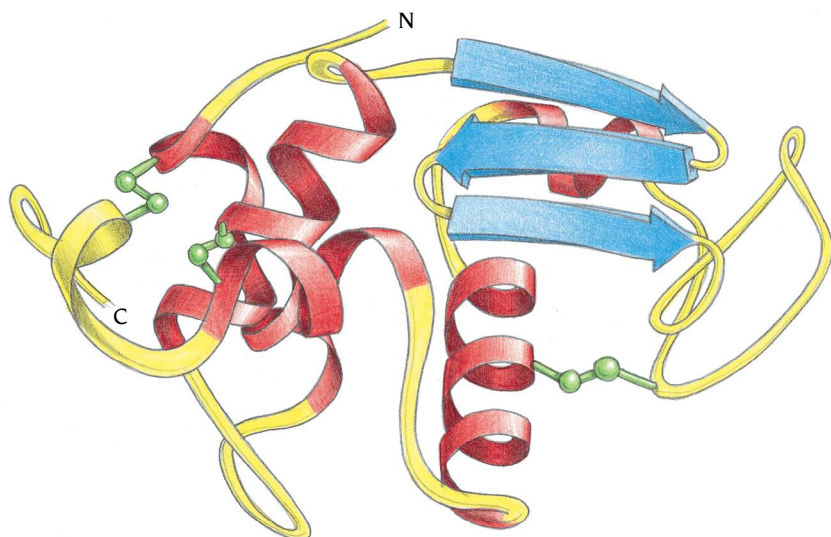


Figure 6.6 Schematic diagram of the structure of the enzyme lysozyme which folds into two domains. One domain is essentially α -helical whereas the second domain comprises a three stranded antiparallel β sheet and two α helices. There are three disulfide bonds (green), two in the α -helical domain and one in the second domain.

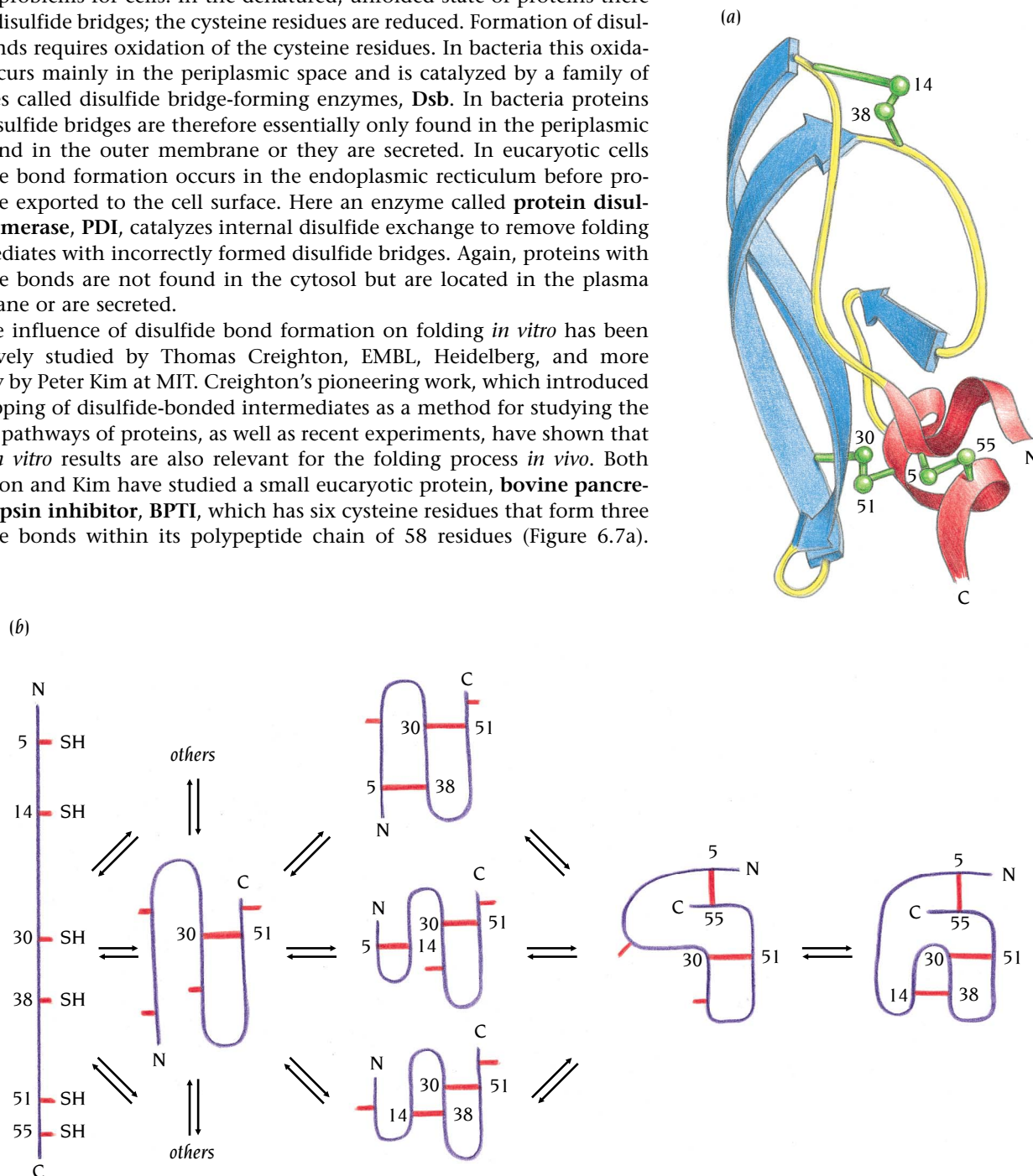
had achieved a high degree of secondary structure while the β -sheet domain contained no detectable structure, and a second state in which no stable structure was observed in either the α or the β domain. In addition, a third, less populated state was observed in which both the α domain and the β domain had significant structure. Therefore in this case it is likely that the folding follows different pathways, two major and one minor, with different molten globule states which in a later stage of the folding process converge to one final native state (see Figure 6.5b).

Enzymes assist formation of proper disulfide bonds during folding

The formation of correct disulfide bonds during the folding process poses special problems for cells. In the denatured, unfolded state of proteins there are no disulfide bridges; the cysteine residues are reduced. Formation of disulfide bonds requires oxidation of the cysteine residues. In bacteria this oxidation occurs mainly in the periplasmic space and is catalyzed by a family of enzymes called disulfide bridge-forming enzymes, **Dsb**. In bacteria proteins with disulfide bridges are therefore essentially only found in the periplasmic space and in the outer membrane or they are secreted. In eucaryotic cells disulfide bond formation occurs in the endoplasmic reticulum before proteins are exported to the cell surface. Here an enzyme called **protein disulfide isomerase, PDI**, catalyzes internal disulfide exchange to remove folding intermediates with incorrectly formed disulfide bridges. Again, proteins with disulfide bonds are not found in the cytosol but are located in the plasma membrane or are secreted.

The influence of disulfide bond formation on folding *in vitro* has been extensively studied by Thomas Creighton, EMBL, Heidelberg, and more recently by Peter Kim at MIT. Creighton's pioneering work, which introduced the trapping of disulfide-bonded intermediates as a method for studying the folding pathways of proteins, as well as recent experiments, have shown that these *in vitro* results are also relevant for the folding process *in vivo*. Both Creighton and Kim have studied a small eucaryotic protein, **bovine pancreatic trypsin inhibitor, BPTI**, which has six cysteine residues that form three disulfide bonds within its polypeptide chain of 58 residues (Figure 6.7a).

Figure 6.7 (a) Schematic diagram of the structure of the small protein bovine pancreatic trypsin inhibitor, BPTI. The three disulfide bonds are green, β strands are blue and α helices are red. (b) The folding pathway of BPTI according to Creighton. The unfolded protein has six cysteine residues in their reduced state. The major single disulfide bond intermediate has a disulfide bond between residues 30 and 51. This intermediate forms double disulfide bond intermediates which contain non-native disulfide bonds. According to Creighton these intermediates are essential for the formation of the native double disulfide bond intermediate 30–51, 5–55 which rapidly forms the third native disulfide bond 14–38.



The fully reduced protein is largely unfolded and does not fold until the cysteine residues are oxidized to disulfide bridges. In the native state these bonds are between cysteine residues 30–51, 5–55 and 14–38. During the folding process formation of the first disulfide bond is almost random and the 15 possible single-disulfide species are in rapidly interchanging equilibrium. However, the intermediate with the disulfide bond at 30–51 is more stable than the other 14 and is present in about 60% of the molecules. The productive folding pathway goes through this stable intermediate, consequently all other intermediates must eventually rearrange their disulfide bond in order to fold properly. It has a partly folded conformation comprising the native-like α helix and β sheet linked by the 30–51 disulfide bond. The remaining part of the polypeptide chain is disordered or unfolded and includes cysteines 5, 14 and 38 but not cysteine 55 which is within the folded α helix.

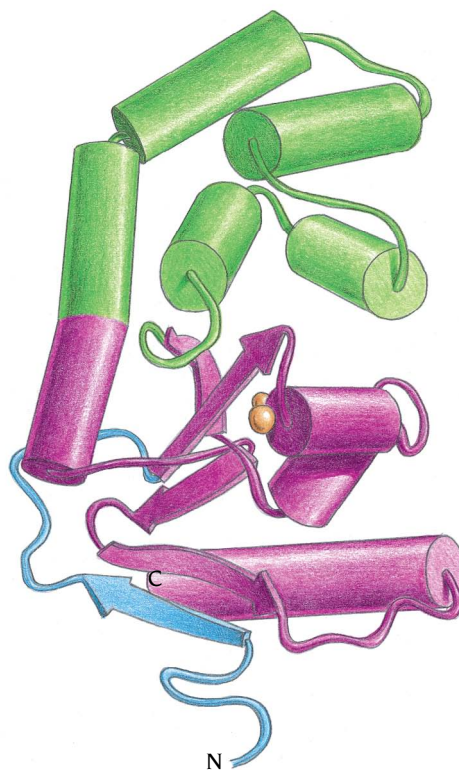
The second disulfide bonds formed in the 30–51 intermediate are between all three possible pairs of flexible cysteines, to form 5–14, 5–38 or 14–38 disulfide bonds (Figure 6.7b). The first two are non-native disulfide bonds, which are not stabilized to any significant extent, and occur primarily because they are in flexible parts of the polypeptide chain. In contrast, upon forming the 14–38 disulfide bond, there are now two native disulfide bonds, and the molecule adopts a very native-like, folded conformation. However, formation of the last disulfide bond between cysteines 5 and 55 occurs only very slowly in this intermediate because these cysteine residues are buried inside the folded intermediate and not accessible to oxidizing agents or disulfide rearrangement. The productive precursor to the native state is instead the intermediate with the 30–51 and 5–55 disulfide bonds since the remaining cysteines 14 and 38 are in good proximity on the surface of the folded protein and there is no barrier to them forming the final disulfide bond. This intermediate can be reached either by rearrangement of the non-native disulfide bonds formed in the 30–51 intermediate or by unfolding the 30–51, 14–38 intermediate.

The folding pathway of BPTI illustrates clearly the importance of disulfide rearrangements for the folding process and hence the advantage for a cell of having enzymes that increase the rate of this reaction. Adding the enzyme protein disulfide isomerase significantly increases the rate of folding of BPTI *in vitro*. The three-dimensional structure of a eucaryotic protein disulfide isomerase has not yet been determined but the structure of DsbA from *Escherichia coli*, a member of the Dsb family of enzymes, has been solved by John Kuriyan, Rockefeller University. This monomeric enzyme has 189 amino acid residues that fold into two domains; one comprises five tightly packed α helices while the second domain has a structure very similar to that of **thioredoxin** (Figure 6.8). Thioredoxin, the x-ray structure of which was determined by the group of Carl Branden, Uppsala, is a ubiquitous protein that functions as a general protein disulfide oxido-reductase and in bacteria is responsible for keeping disulfide bridges reduced. The well-characterized mechanism of thioredoxin is based on reversible oxidation of two cysteine thiol groups to a disulfide, accompanied by the transfer of two electrons and two protons. Presumably DsbA functions in a similar way since the redox-active disulfide bridges in these two proteins are very similar.

The thioredoxin domain (see Figure 2.7) has a central β sheet surrounded by α helices. The active part of the molecule is a $\beta\alpha\beta$ unit comprising β strands 2 and 3 joined by α helix 2. The redox-active disulfide bridge is at the amino end of this α helix and is formed by a Cys-X-X-Cys motif where X is any residue in DsbA, in thioredoxin, and in other members of this family of redox-active proteins. The α -helical domain of DsbA is positioned so that this disulfide bridge is at the center of a relatively extensive hydrophobic protein surface. Since disulfide bonds in proteins are usually buried in a hydrophobic environment, this hydrophobic surface in DsbA could provide an interaction area for exposed hydrophobic patches on partially folded protein substrates.

Disulfide bonds in proteins are generally stable and nonreactive, acting like bolts in the structure. However, oxidized DsbA is less stable than the reduced form and its disulfide bond is very reactive. DsbA is thus a strong

Figure 6.8 Schematic diagram of the enzyme DsbA which catalyzes disulfide bond formation and rearrangement. The enzyme is folded into two domains, one domain comprising five α helices (green) and a second domain which has a structure similar to the disulfide-containing redox protein thioredoxin (violet). The N-terminal extension (blue) is not present in thioredoxin. (Adapted from J.L. Martin et al., *Nature* 365: 464–468, 1993.)



oxidizing agent which is consistent with its ability to catalyze the exchange of disulfide bonds in other proteins. Although the structure of eucaryotic protein disulfide isomerase is unknown, sequence alignment shows that its 189 amino acid polypeptide chain contains two domains that are homologous to thioredoxin. This indicates that both the bacterial and the eucaryotic proteins that catalyze the formation of disulfide bonds in proteins during folding use thioredoxin-like domains and presumably are evolutionarily related.

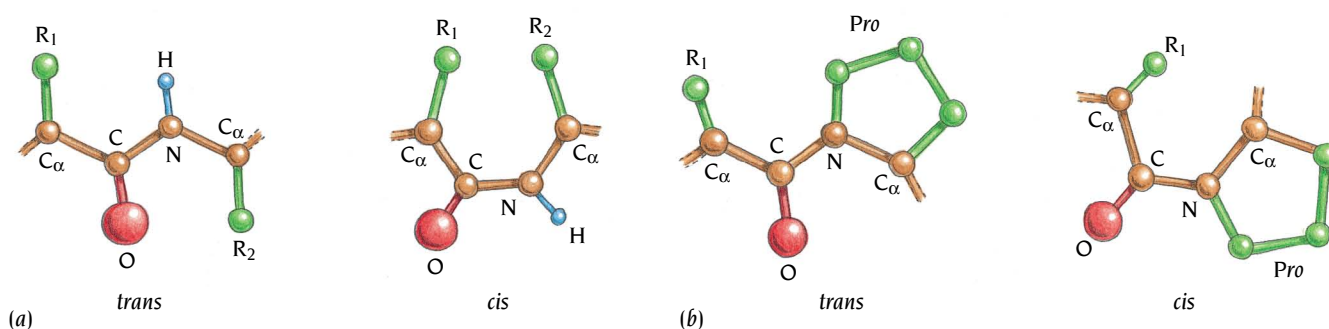
Isomerization of proline residues can be a rate-limiting step in protein folding

In Chapter 1 we presented a picture of the peptide as a planar unit with the C=O and N-H groups pointing in opposite directions in the plane (see Figure 1.5). This so called *trans*-peptide is the most stable form of the peptide group. However, there is another possible form, the *cis*-peptide, which is also planar but in which the C=O and N-H groups point in the same direction (Figure 6.9a). For most peptides the *cis*-form is about 1000 times less stable than the *trans*-form and consequently *cis*-peptides are rarely found in native proteins. However, when the second residue (R_2 in Figure 6.9a) is a proline the *cis*-form is only about four times less stable than the *trans*-form (Figure 6.9b) and some *cis*-proline peptides occur in many proteins. Most proline residues in proteins are of course in the *trans*-configuration because in *trans* there are fewer steric collisions, but *cis*-prolines are found in tight bends of the polypeptide chain and are sometimes essential for activity or for conformational flexibility.

In the native protein these less stable *cis*-proline peptides are stabilized by the tertiary structure but in the unfolded state these constraints are relaxed and there is an equilibrium between *cis*- and *trans*-isomers at each peptide bond. When the protein is refolded a substantial fraction of the molecules have one or more proline-peptide bonds in the incorrect form and the greater the number of proline residues the greater the fraction of such molecules. *Cis-trans* isomerization of proline peptides is intrinsically a slow process and *in vitro* it is frequently the rate-limiting step in folding for those molecules that have been trapped in a folding intermediate with the wrong isomer.

In vivo the rate of this process is enhanced by enzymes initially called **peptidyl prolyl isomerases**, which are found in both procaryotic and eucaryotic organisms. Surprisingly these enzymes were later found to be involved in immunosuppression by inhibiting T cell proliferation after binding of immunosuppressive drugs; this medically important activity of peptidyl prolyl isomerases is unrelated to their isomerase activity in folding. The first peptidyl prolyl isomerase to be discovered is now called **cyclophilin** because of its role as target for the most frequently used therapeutic agent for prevention of graft rejection, cyclosporin A. Cyclophilin is an abundant soluble protein consisting of a single polypeptide chain of 165 amino acids. Because it is a target for cyclosporin A the determination of its structure was pursued in the early 1990s by several pharmaceutical companies intent

Figure 6.9 (a) Peptide units can adopt two different conformations, *trans* and *cis*. In the *trans*-form the C=O and the N-H groups point in opposite directions whereas in the *cis*-form they point in the same direction. For most peptides the *trans*-form is about 1000 times more stable than the *cis*-form. (b) When the second residue in a peptide is proline the *trans*-form is only about four times more stable than the *cis*-form. *Cis*-proline peptides are found in many proteins.



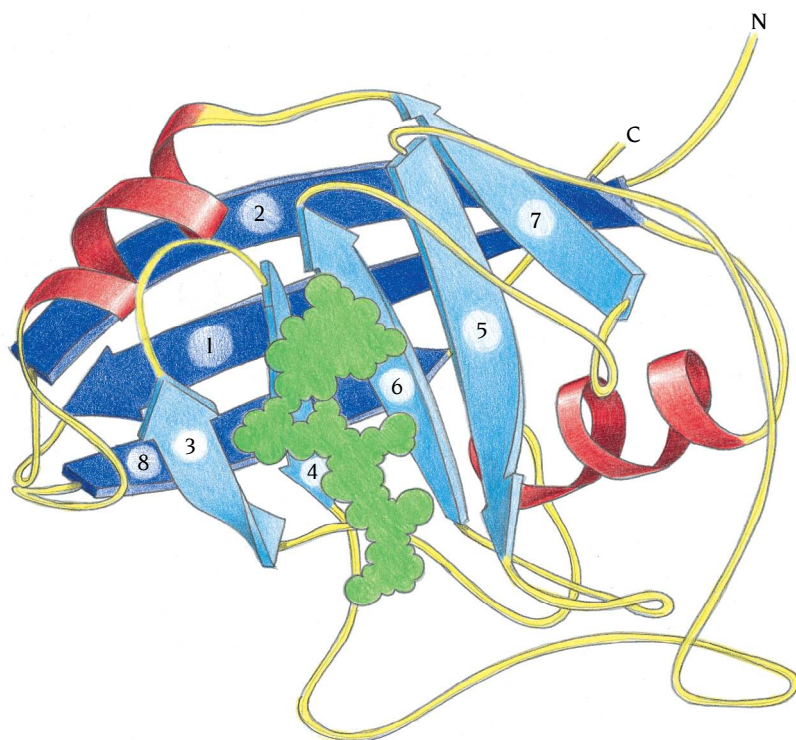


Figure 6.10 Schematic diagram of the structure of the protein cyclophilin, a prolyl peptide isomerase that catalyzes the conversion between *cis*- and *trans*-proline peptides. The protein is folded into an eight-stranded antiparallel β barrel (blue) with two α helices (red) on the outside. The active site, which has been located by the binding of a proline containing tetrapeptide (green), is on the outside of the β barrel. (Adapted from J. Kallen et al., *Nature* 353: 276–279, 1991.)

upon designing better immunosuppressive drugs, and a group from Sandoz, Basel, in collaboration with Kurt Wüthrich, ETH, Zurich, published the first structure of cyclophilin with a bound proline-containing peptide using a combination of x-ray crystallography and NMR.

Their work showed that the structure of cyclophilin complexed with a proline-containing tetrapeptide (Figure 6.10) is essentially an eight-stranded β barrel with an unusual topology, different from both the up-and-down and the Greek key β barrels discussed in Chapter 5. The interior of the barrel is tightly packed with hydrophobic residues. In contrast to the up-and-down β barrels that bind ligands inside the barrel, the tetrapeptide ligand of cyclophilin is bound on the outside of the barrel in a deep groove between one face of the β sheet and a long loop region (see Figure 6.10).

Cyclophilin enhances the rate of *cis-trans* isomerization of proline peptides by a factor of one million over the nonenzymatic rate. The detailed mechanism of the enzymatic process is still unclear but mechanisms based on distortion or desolvation of the peptide group are possible. The proline residue is tightly bound in a hydrophobic pocket of the binding groove and the carbonyl oxygen atom is hydrogen-bonded to basic side chains. The binding of a peptide segment into a hydrophobic environment may promote *cis-trans* isomerization by decreasing the charge separation in the peptide group (see Figure 2.3a) and thus creating a peptide bond with a more single-bond character. Alternatively, the tight hydrophobic interactions with cyclophilin might distort the geometry of the peptide group toward the transition state for isomerization. Both mechanisms or a combination of them will decrease the energy barrier for rotation around the peptide bond, which is required for the *cis-trans* isomerization.

Proteins can fold or unfold inside chaperonins

Before protein molecules attain their native folded state they may expose hydrophobic patches to the solvent. Isolated purified proteins will aggregate during folding even at relatively low protein concentrations. Inside cells, where there are high concentrations of many different proteins, aggregation could therefore occur during the folding process. This is prevented by

molecular chaperones, ubiquitous and abundant families of proteins that assist the folding of both nascent polypeptides still attached to ribosomes and released completed polypeptide chains. Several chaperones are induced by heat shock, because protein unfolding and aggregation is increased at elevated temperatures; and consequently chaperones were first classified as **heat-shock proteins** (Hsps). Several of these heat-shock proteins, such as the 70 kDa protein **DnaK** (**Hsp 70**) are also expressed normally and participate in various other cellular processes including assembly and disassembly of multimeric proteins and membrane translocation of secreted proteins. The Hsp 70 polypeptide chain is divided into two functional regions, one that binds and hydrolyses ATP and a second that binds hydrophobic segments of unfolded polypeptide chains. The N-terminal ATP-binding region has a four-domain structure very similar to that of actin (Chapter 14); the polypeptide-binding domain is an antiparallel β sandwich in the C-terminal region. There is as yet no structural information on the complete Hsp 70 molecule. By contrast, the role during folding of two other heat-shock proteins, Hsp 60 and Hsp 10, with molecular weights of 60 kDa and 10 kDa, respectively, has been studied extensively, especially in *Escherichia coli* where they are called **GroEL** and **GroES**, respectively. These proteins function together as a large complex, called a **chaperonin**, consisting of 14 subunits of GroEL and 7 subunits of GroES and requiring ATP to function.

Chaperonins bind unfolded, partly folded and incorrectly folded protein molecules but not proteins in their native state. They are promiscuous in that they bind to and assist the folding of a large number of different proteins independent of the latter's amino acid sequences. How can these chaperonins distinguish between correctly and incorrectly folded versions of almost any water-soluble polypeptide chain and how can they mediate the efficient conversion of unfolded or misfolded proteins to their native form? The x-ray structure determinations of GroEL and GroES, in combination with electron microscopic studies of GroEL–GroES–polypeptide complexes, have made possible major steps towards understanding these processes.

GroEL is a cylindrical structure with a central channel in which newly synthesized polypeptides bind

The x-ray structure of GroEL was determined in 1994 by the groups of Paul Sigler and Arthur Horwich, Yale University. The 14 subunits of GroEL, each comprising 547 amino acids, form two rings in which the 7 subunits of each ring are arranged with nearly exact sevenfold rotational symmetry. The rings are arranged back-to-back, forming an extensive interface with one another across an almost flat equatorial plane. The whole structure resembles a porous thick-walled cylinder that is about 150 Å long and 140 Å in diameter and contains a large central cavity or channel (Figure 6.11). Each subunit has three distinct domains, equatorial, intermediate and apical, that are arranged in the cylinder as shown in Figure 6.12. The equatorial domain which is largest and comprises 243 residues is mainly α -helical. It serves as the foundation of the GroEL structure providing all of the contacts between the two sevenmembered rings across the equatorial plane. In addition it provides most of the contacts between the subunits within each ring as well as the ATP binding site, which is essential for function. The equatorial domain contains both the N-terminus and the C-terminus of the polypeptide chain. About 30 residues at these ends are not visible in the x-ray structure; either they are disordered, or they occupy differently ordered positions in different molecules. The visible ends point into the central cavity close to the equatorial plane and presumably the nonvisible residues partially block the channel in this region; three-dimensional electron microscopic reconstruction of individual chaperonin particles shows that the channel is narrow in the center of the molecule (Figure 6.13).

The apical domain (residues 191–376) is essentially a four-layer structure comprising two β sheets sandwiched between α helices. One β sheet has

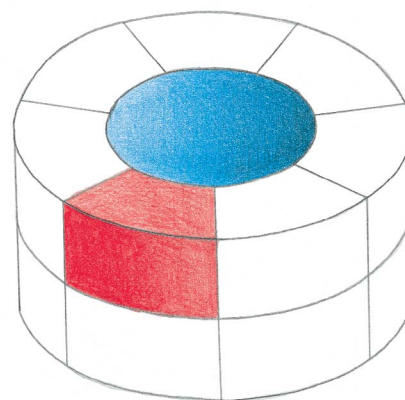


Figure 6.11 Schematic diagram of the chaperonin GroEL molecule as a cylinder with 14 subunits arranged in two rings of 7 subunits each. The space occupied by one subunit is red and the hole inside the cylinder is blue.

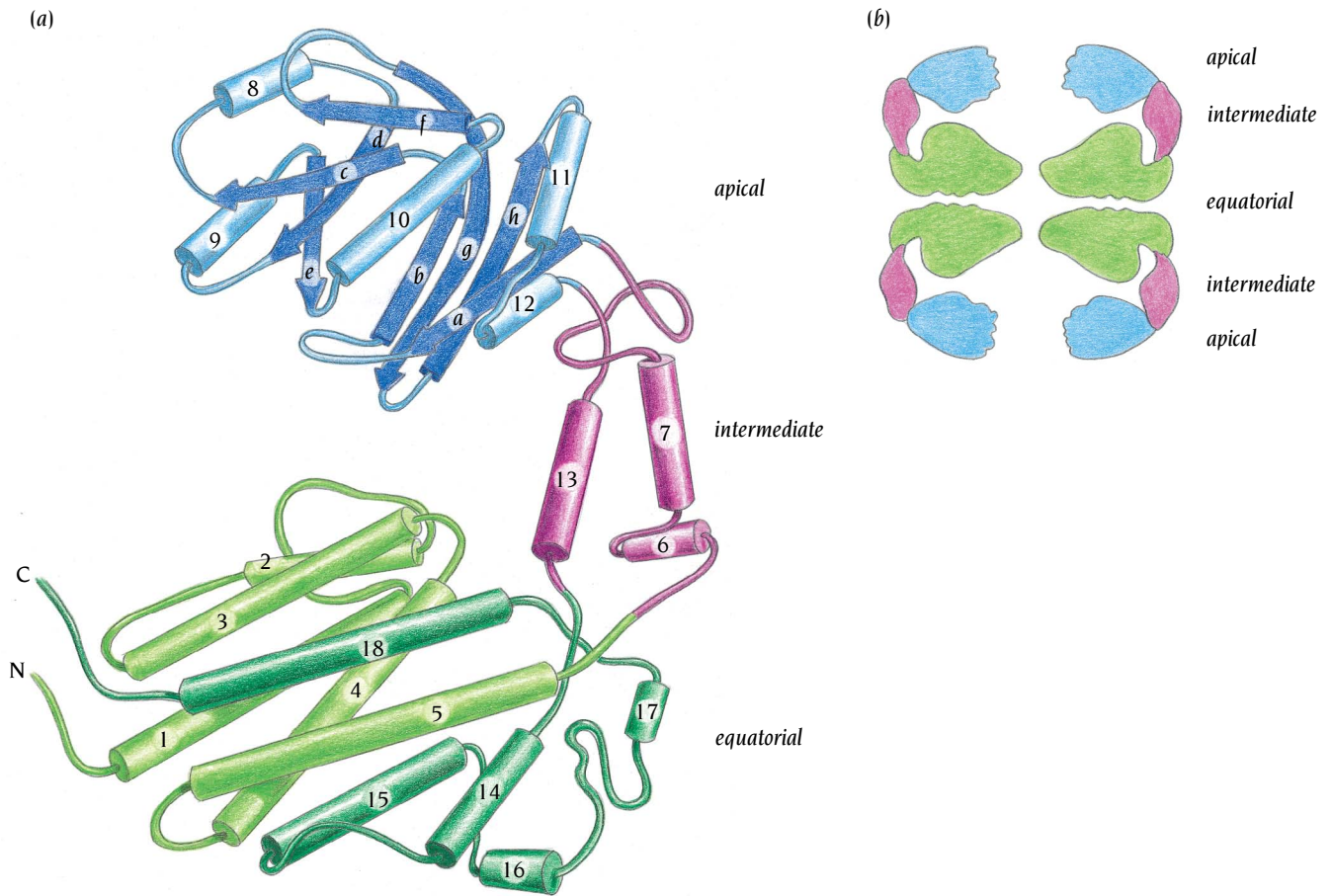


Figure 6.12 (a) Schematic diagram of one subunit of GroEL. The polypeptide chain is folded into three domains. The equatorial domain (green) is the largest domain, comprising 10 α -helices, and is built up from both the N-terminal and the C-terminal regions. The apical domain (blue), which is a β -sandwich flanked by α -helices, is formed by the middle region of the polypeptide chain. The two linker regions between the equatorial and the apical domains form a small intermediate domain (purple) comprising three α -helices. (b) Schematic diagram illustrating the domain arrangement of four subunits in the GroEL molecule, two in each of the seven membered rings. The equatorial domains form the middle part of the molecule and interact with each other both within each ring and between the rings. The apical domains are at the top and the bottom of the cylinder and form an opening to the interior of the molecule. The small intermediate domains form the thinnest part of the cylinder wall in the middle of each ring. [(a) Adapted from K. Braig et al., *Nature* 371: 578–586, 1994.]

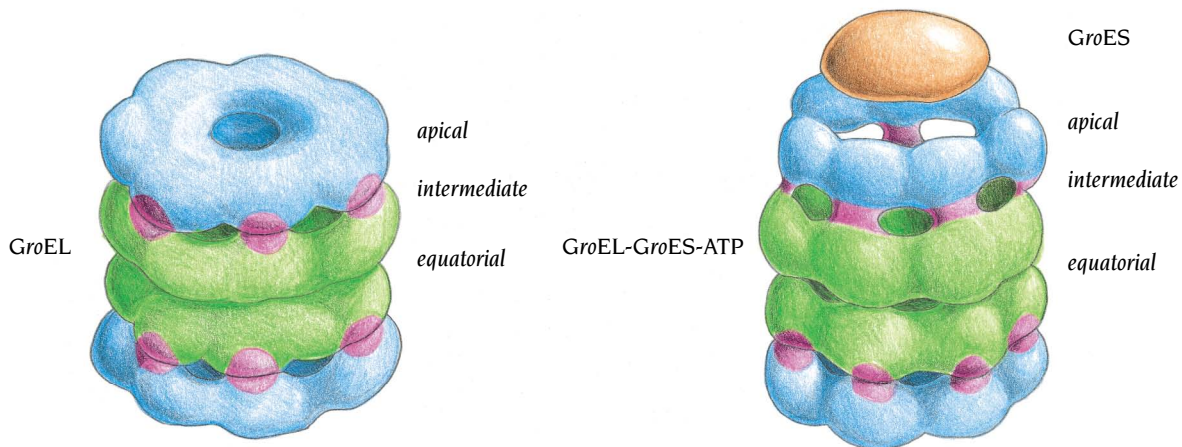


Figure 6.13 Models of the GroEL molecule in two different functional states based on three-dimensional reconstruction from electron microscopy pictures. A large conformational change of GroEL occurs when GroES and ATP are bound. The GroES molecule binds to one of the GroEL rings and closes off the central cavity. The GroEL ring becomes larger and the cavity inside that part of the cylinder becomes wider. (Adapted from S. Chen et al., *Nature* 371: 261–264, 1994.)

antiparallel β strands whereas the other is part of an α/β domain with four parallel β strands. The apical domains form the opening to the solvent of the central channel. The segments of this domain that form the top surface of the molecule as well as those that face the upper regions of the channel are flexible and not very well ordered. These segments are rich in hydrophobic residues and they are involved in binding to the hydrophobic areas exposed by non-native folds of polypeptide chains. Mutating these hydrophobic residues to charged ones abolishes both GroES and polypeptide binding whereas mutations conserving hydrophobic residues, such as Phe to Val, have no functional effects. The flexibility of these segments probably accounts for the promiscuous binding of a wide range of unfolded polypeptide sequences.

The equatorial and apical domains are linked by a small intermediate domain which forms part of the outer wall and extends only about 25 Å in the radial direction; consequently the internal cavity is wider in this region, up to 90 Å in diameter. In addition there are holes in the wall between the intermediate domains in adjacent subunits. These seven holes are large enough to permit ATP and ADP to diffuse in and out. The intermediate domain is connected to the other two domains through short antiparallel segments that could easily serve as hinges during conformational changes. Electron microscopic studies of GroEL with different ligands bound have shown that substantial changes in the orientations of the domains and in the size of the central cavity occur during the functional cycle of the chaperonin (see Figure 6.13).

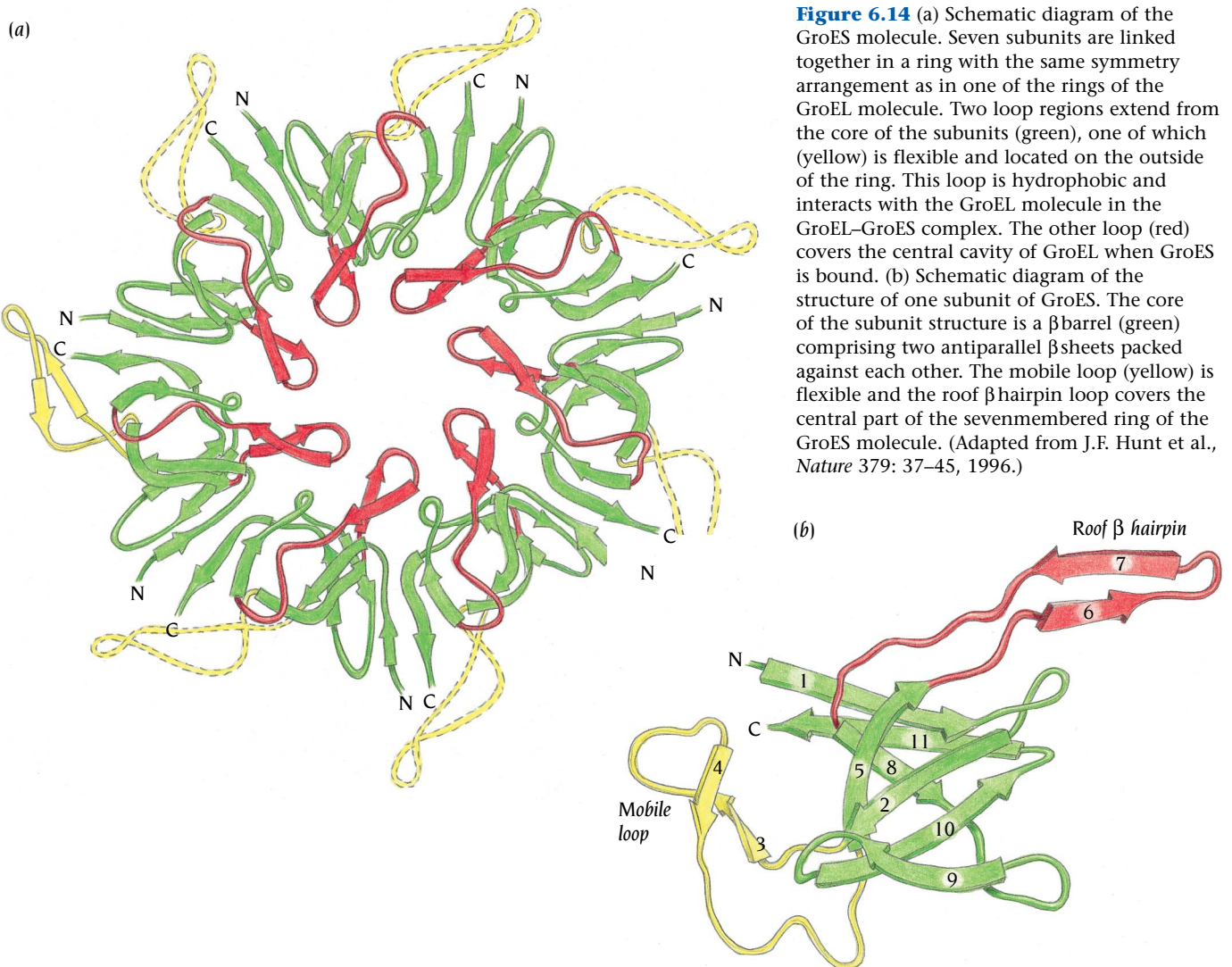
GroES closes off one end of the GroEL cylinder

GroES binds to the apical domain of GroEL, closing off the central cavity (see Figure 6.13). Once GroES has bound to one of the rings in the GroEL molecule a conformational change occurs which decreases the affinity of the second GroEL ring for GroES. The predominant functional state of the GroEL–GroES complex is, therefore, asymmetric with GroES bound to only one end of the GroEL cylinder. Obviously there is strong structural communication between the halves of the GroEL molecule since GroES binding to one half affects the properties of the other half, despite a distance of about 150 Å between the two GroES binding sites.

The GroES molecule comprises seven subunits, each of 97 amino acids linked together around a sevenfold rotation axis, the same symmetry arrangement as in one GroEL ring. The x-ray structure of GroES was determined in 1996 by the group of Johann Deisenhofer, University of Texas, Dallas. The GroES molecule is shaped like a dome, about 75 Å in diameter and 30 Å high with a small hole in the middle (Figure 6.14a). The core of the subunit structure is a β barrel comprising two antiparallel β sheets packed against each other (Figure 6.14b). Two large loop regions protrude from this core, one of which extends above the plane of the ring creating a loosely packed top of the dome that covers the central hole. The other loop region, which is rich in hydrophobic residues, extends below the dome and presumably interacts with the apical domain in the GroES–GroEL complex. This loop is disordered in the x-ray structure of GroES but NMR studies have shown that the loop becomes ordered when GroES binds to GroEL and mutational studies have shown that the hydrophobic residues in this loop are important for chaperonin function.

The GroEL–GroES complex binds and releases newly synthesized polypeptides in an ATP-dependent cycle

How does the GroEL–GroES complex function as a chaperone to assist protein folding? Although several aspects of the mechanism are not clear, the main features of the functional cycle are known. The first step is the



formation of a GroEL–ATP complex, one end of which which then binds one molecule of GroES, with the hydrolysis of ATP. The resulting GroEL–ADP–GroES is a stable complex the halves of which have very different properties (Figure 6.15a). The GroEL ring where GroES is bound (*cis*-position) has undergone a large structural change forming a wide internal cavity whose walls are formed from both the apical and intermediate domains. This cavity is partly closed off from the solvent by the GroES dome. The other ring (*trans*-position) has a smaller cavity which is open to the solvent. Unfolded proteins can bind in both the *cis*- and the *trans*-positions but only those that are bound in the *cis*-position undergo subsequent folding rearrangements (Figure 6.15c–e). Binding and release of polypeptides at the *trans*-position seem to be functionally unimportant.

Release of bound polypeptides from the closed cavity in the *cis*-position requires that GroES is first released. The release of GroES, like its binding, requires ATP hydrolysis, but this time by ATP molecules bound to the distant GroEL ring in the *trans*-position (Figure 6.15e,f). This is another example of the strong structural communication between the halves of the complex since ATP hydrolysis in the *trans*-GroEL ring affects the GroES binding site almost 100 Å away. Once GroES is released the polypeptide chain is released and it can bind to another GroEL–GroES complex to repeat the cycle. The native state is reached after iterative rounds involving multiple binding and release to GroEL–GroES complex.

The crucial question that remains to be answered is, of course, what happens to the polypeptide chain inside the closed *cis*-cavity. Two different

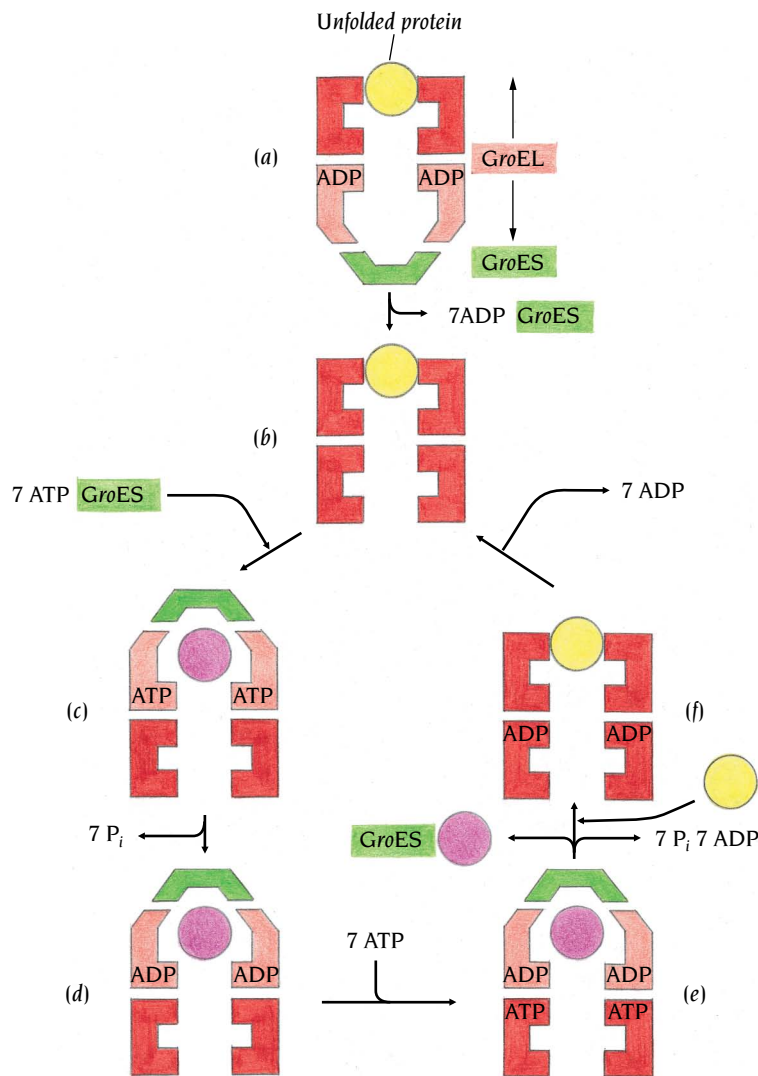


Figure 6.15 Possible functional cycle of the GroEL-GroES molecule. (a) An unfolded protein molecule (yellow) binds to one end of the GroEL-ADP complex (red) with bound GroES (green) at the other end. (b and c) GroES is released from the *trans*-position and rebound together with ATP at the *cis*-position (light red) of GroEL. (d) ATP hydrolysis occurs as the protein is folding or unfolding inside the central cavity. (e) ATP binding and hydrolysis in the *trans*-position is required for release of GroES and the protein molecule. (f) A new unfolded protein molecule can now bind to GroEL. (Adapted from M. Mayhew et al., *Nature* 379: 420–426, 1996.)

models have been proposed. Both envisage that unfolded or incorrectly folded proteins are recognized by their exposed hydrophobic areas and bound to hydrophobic regions inside the GroEL cavity. In one of these models the subsequent function of the cavity is to unfold unproductive intermediates and then eject them in the unfolded state into the bulk solution for spontaneous folding, giving them another chance to reach the folded state. In this model folding would occur in the solvent during jumps of the polypeptide between GroEL-GroES complexes. In the second model folding occurs inside the *cis*-cavity of the GroEL-GroES complex, either to the native state or to an intermediate state along one of the productive pathways to the native state. Experiments by the group of Ulrich Hartl, Memorial Sloan-Kettering Cancer Center, New York, have provided support, at least for some proteins, for the second model, which has the attractive feature that folding occurs in a closed environment preventing aggregation with other unfolded proteins during the folding process. However, GroEL-assisted folding of large protein molecules must be different since they are too large to fit inside the cavity. In addition, assisted folding by Hsp 70 does not occur inside a closed cavity.

The folded state has a flexible structure

For simplicity we have so far described a native folded protein molecule as being in one single state. However, within this state, the protein molecule does not have a static rigid structure at normal temperatures. Instead, all the atoms are subject to small temperature-dependent fluctuations. The molecule

as a whole undergoes **breathing** and every atom is constantly in motion. These atomic movements are in general random, but sometimes they can be collective and cause groups of atoms to move in the same direction. Side chains can flip from one conformation to another, some loop regions may not be fixed in one single conformation, helices may slide relative to each other and entire domains can change their packing contacts and open or close the distance between them. Usually these motions are small, a few tenths of an Ångström; but occasionally the collective motions can be large and very significant.

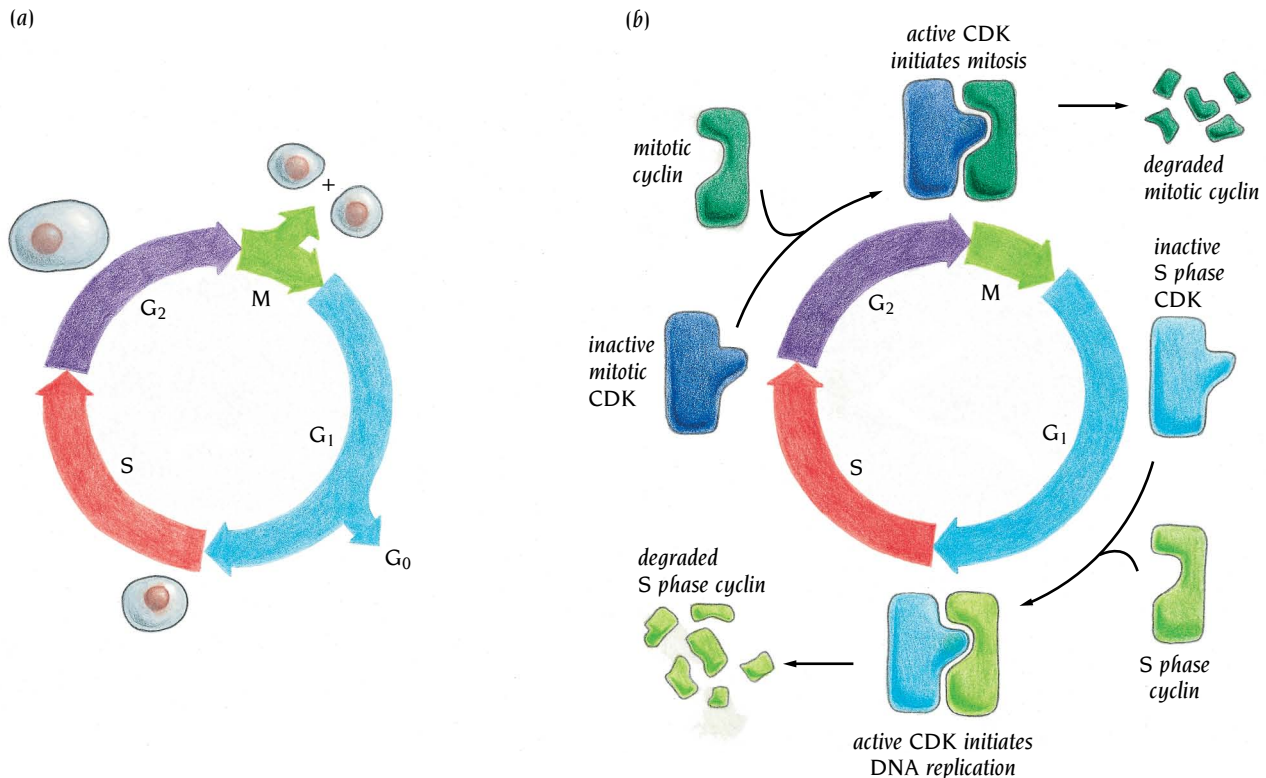
Such large collective movements are reflected in x-ray studies as a low level of electron density and in some cases no electron density at all. The regions that undergo these movements are usually described by crystallographers as flexible or disordered, but in order to distinguish between collective movements and a few discrete and well-ordered but different conformations of these regions, x-ray studies must be made at different temperatures. In NMR studies the experimental data for these regions are compatible with many different conformations. Insight into these individual and collective motions has been obtained by theoretical studies, called molecular dynamics simulations, which use classical Newtonian descriptions of atomic movements.

These calculations have shown that collective movements occur on the picosecond time scale for individual residues, and in nanoseconds for loop regions. Such movements are very important for the function of many protein molecules. Reactions such as electron transfer or ligand binding and release occur on these time scales and usually require movements of protein atoms. As soon as the structure of myoglobin was determined it was immediately apparent that the static picture of the myoglobin molecule seen in the crystal did not allow oxygen atoms to enter its binding site or diffuse out. We now know that while the myoglobin molecule is breathing, pathways are opened up between the solvent and the buried binding site to allow oxygen binding or release on a time scale of nanoseconds.

In addition to these small breathing movements of protein atoms there can also be larger conformational changes between different functional states of the molecule. Small differences in the environment such as different pH or the presence or absence of ligands can stabilize different conformational states of the protein. These conformational changes can vary from adjustments of side chain orientations in the active site to movements of loop regions, differences in relative orientation of domains or changes in the quaternary structure of oligomeric proteins. Such movements are usually essential for function, for enzyme catalysis, binding of antigens to antibodies, receptor–ligand interactions, muscle action and energy transduction and so on. We have already discussed two extreme examples, the effect of pH change on hemagglutinin (Chapter 5) and ATP hydrolysis in GroEL–GroES. We will here give a few further examples of striking conformational transitions.

Conformational changes in a protein kinase are important for cell cycle regulation

The cell division cycle of all eucaryotic cells can be divided into four major phases: G₁, S, G₂, and M (Figure 6.16a). In S phase (DNA synthesis), the entire DNA content of the nucleus is duplicated and the number of chromosomes doubled; in M phase (mitosis), the duplicated chromosomes of the parental cell are segregated using the microtubules of the mitotic spindle, such that each daughter cell receives the same set of chromosomes. After mitosis, including cytokinesis which divides the cytoplasm between the two daughter cells, and before the onset of the next S phase there is an interval designated Gap 1 or G₁. After the completion of DNA replication and chromosome duplication, in other words after S phase is completed, there is a second interval designated Gap 2, or G₂, before the onset of mitosis. The



complete cycle therefore is M, G₁, S, G₂ and M again. Throughout G₁, S and G₂ the cell's protein-synthesising machinery, macromolecules and organelles are built up, and the cell increases in volume. During mitosis the chromosomes and the cytoplasm are divided into two essentially equal parts. An additional stationary phase G₀ occurs in cells which are not actively dividing. How is the progression of a dividing cell through the consecutive phases of the cell cycle regulated? What triggers DNA synthesis in a G₁ cell and mitosis in a G₂ cell? Why don't the two daughter cells emerging from mitosis immediately begin a new round of DNA synthesis?

A combination of biochemical and genetic studies have shown that the progression through the cell cycle is dependent upon the successive activation of a series of enzymes called **cyclin-dependent protein kinases, CDKs** (Figure 6.16b). Each CDK during its transient existence phosphorylates target proteins that then directly or indirectly activate the next set of events of the cell cycle. Each CDK is a heterodimer comprising a catalytic subunit, the protein kinase, complexed with a **cyclin** molecule which activates the kinase. In vertebrate cells there are at least four different CDKs involved in control of passage through the cell cycle, as well as other CDKs with other functions. The different catalytic subunits are products of the genes of a closely related gene family. The different cyclins, one or more for each sort of catalytic subunit, are also members of a gene family. The cyclin components of the CDKs undergo sequential programmed synthesis, accumulation and then degradation; the short half-lives of the cyclins ensure that the CDKs of which they are part are active kinases only for short periods and at the correct time in the cell cycle. The CDKs can therefore act as a relay of switches, governing passage from G₁ to S phase, from G₂ to M phase and all other steps that constitute the cell cycle.

Although there is still a very great deal to learn about the physiological substrates of the CDKs and how the enzymes locate their targets, detailed structural information, including the activation of the kinase by cyclin, is available for CDK2-cyclin A, which regulates DNA replication in human cells. The x-ray structure of a functional fragment of cyclin A was determined by the group of Louise Johnson, Oxford University, that of inactive CDK2 by the group of Sung-Ho Kim, University of California, Berkeley, and that of the

Figure 6.16 (a) The five phases of a standard eucaryotic cell cycle. During M phase growth stops and the cell then divides. DNA replication is confined to the S phase; G₁ phase is the gap between M phase and S phase; G₂ phase is the gap between S phase and M phase. Cells which are not dividing enter the stationary phase, G₀. (b) The regulation of CDKs by cyclin degradation. Only two types of cyclin-CDK complexes are shown, one that triggers S phase and one that triggers M phase. In both cases the activation of CDK requires cyclin binding and its inactivation depends on cyclin degradation. (Adapted from B. Alberts et al, *Essential Cell Biology*, New York: Garland Publishing, 1998.)

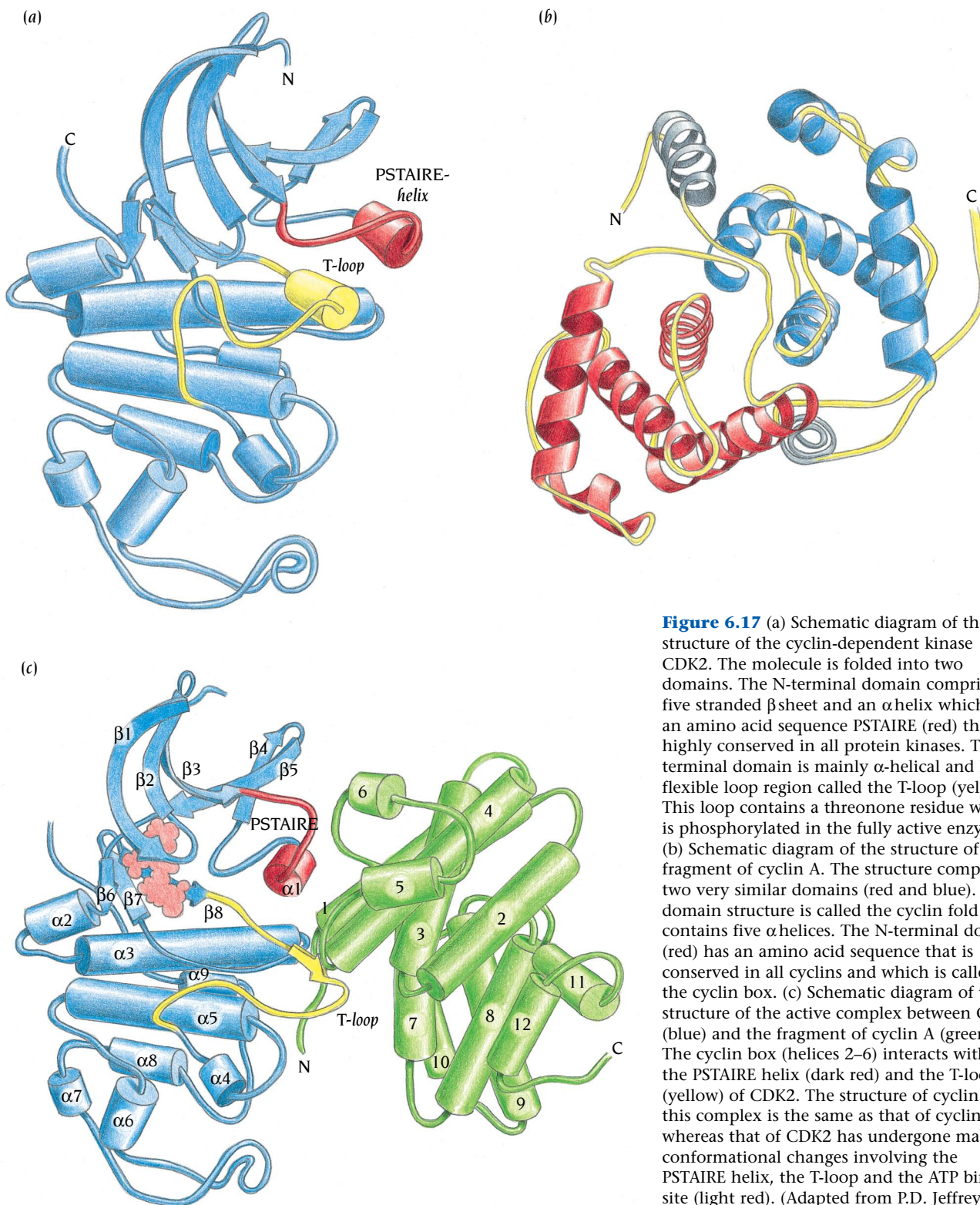


Figure 6.17 (a) Schematic diagram of the structure of the cyclin-dependent kinase CDK2. The molecule is folded into two domains. The N-terminal domain comprises a five stranded β sheet and an α helix which has an amino acid sequence PSTAIRE (red) that is highly conserved in all protein kinases. The C-terminal domain is mainly α -helical and has a flexible loop region called the T-loop (yellow). This loop contains a threonine residue which is phosphorylated in the fully active enzyme. (b) Schematic diagram of the structure of a fragment of cyclin A. The structure comprises two very similar domains (red and blue). This domain structure is called the cyclin fold and contains five α helices. The N-terminal domain (red) has an amino acid sequence that is conserved in all cyclins and which is called the cyclin box. (c) Schematic diagram of the structure of the active complex between CDK2 (blue) and the fragment of cyclin A (green). The cyclin box (helices 2–6) interacts with the PSTAIRE helix (dark red) and the T-loop (yellow) of CDK2. The structure of cyclin A in this complex is the same as that of cyclin A whereas that of CDK2 has undergone major conformational changes involving the PSTAIRE helix, the T-loop and the ATP binding site (light red). (Adapted from P.D. Jeffrey et al., *Nature* 376: 313–320, 1995.)

active cyclin A fragment-CDK2 complex by the group of Nicola Pavletich, Memorial Sloan-Kettering Cancer Center, New York. Comparison of these structures reveals how cyclin A binding to CDK2 causes large conformational changes in the active site of CDK2, converting the protein from an inactive to an active kinase. The structure of cyclin A, in contrast, is not changed.

CDK2 has two domains, a small (85 residue) amino-terminal domain comprising a single α helix and a five-stranded β sheet and a larger (213 residues) domain that is mainly α -helical (Figure 6.17a). The cofactor in the

phosphorylating reaction, ATP, is bound in a cleft between the two domains. The α helix in the small domain contains a sequence of residues, -Pro-Ser-Thr-Ala-Ile-Arg-Glu-, which is highly conserved in all protein kinases and which is called the PSTAIRE helix. Mutational studies have shown that the Glu residue in the PSTAIRE helix plays a crucial role for activity of the enzyme. The large, mainly α -helical domain has a flexible loop region, called the T-loop, which contains a threonine residue, Thr 160 in human CDK2, that is phosphorylated in the fully active enzyme. The cyclin A fragment (residues 173–432) that was used in these x-ray studies is built up from two domains with a very similar structure that is now called the cyclin fold (Figure 6.17b) and which comprises five α helices. The fragment activates CDK2 almost as well as the complete cyclin A molecule. The first domain has an amino acid sequence that is strongly conserved in all cyclins and which is called the cyclin box. In spite of the almost identical structures of the two domains their amino acid sequences are not similar, only one has the cyclin box.

In the cyclin A–CDK2 complex the cyclin box domain interacts with CDK2, mainly with the PSTAIRE helix and the T-loop (Figure 6.17c). The structure of cyclin A in the complex is virtually the same as that of cyclin A alone whereas that of CDK2 has undergone major conformational changes. The whole of the N-terminal domain has slightly changed its orientation relative to the C-terminal domain. In addition the PSTAIRE helix has moved closer to the active site cleft of CDK2 and rotated about 90° so that the catalytically essential residue Glu 51 points into the cleft instead of away from it in free CDK2 (Figure 6.18). Some of the main chain atoms of this helix have moved up to 8 Å due to these concerted movements. Coupled to the structural change of the PSTAIRE helix there is a major rearrangement of the T-loop with some residues moving up to 20 Å (Figure 6.19). As well as adopting a completely new position, the T loop undergoes a structural transformation. The part of the T-loop that is α -helical in free CDK2 melts away, and instead a β strand appears in the complex.

In free CDK2 the active site cleft is blocked by the T-loop and Thr 160 is buried (Figure 6.20a). Substrates cannot bind and Thr 160 cannot be phosphorylated; consequently free CDK2 is inactive. The conformational changes induced by cyclin A binding not only expose the active site cleft so that ATP and protein substrates can bind but also rearrange essential active site residues to make the enzyme catalytically competent (Figure 6.20b). In addition Thr

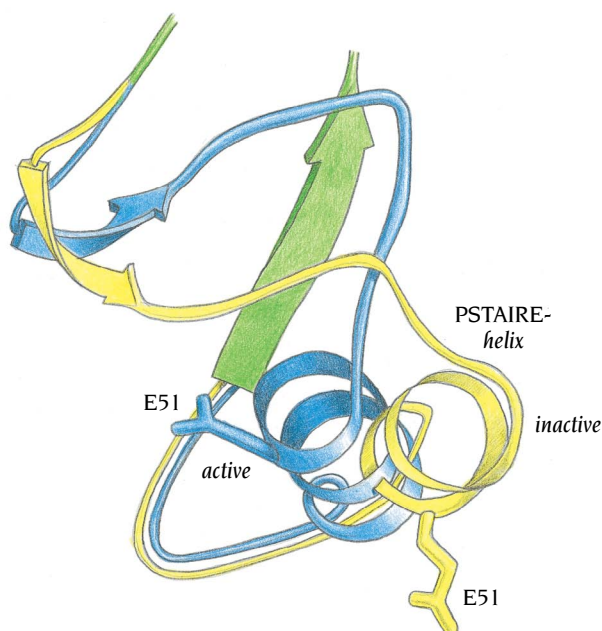
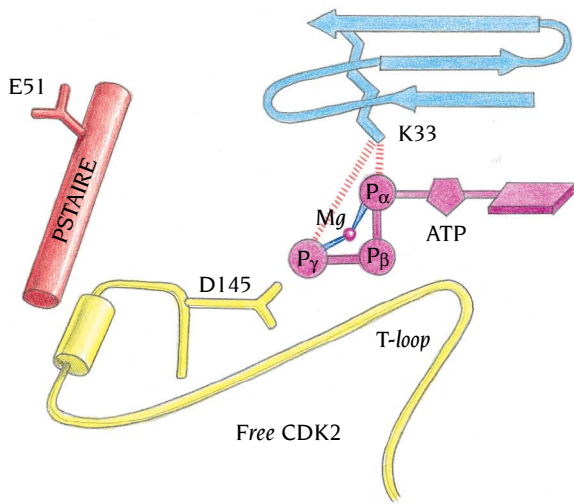
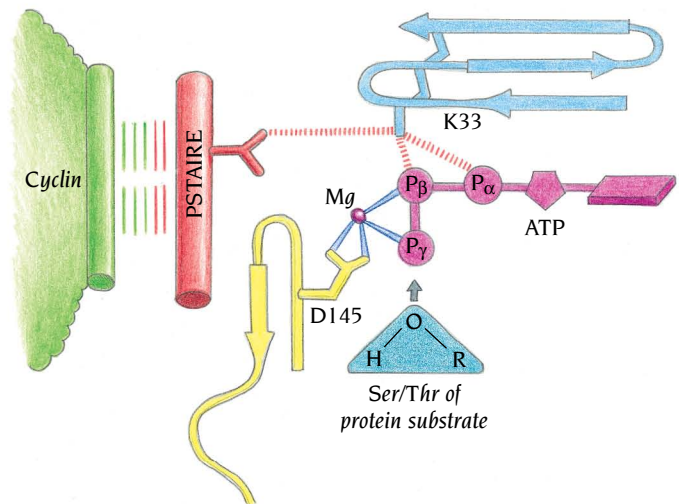


Figure 6.18 The PSTAIRE helix undergoes a major conformational change when CDK2 binds to cyclin A. In the inactive free CDK2 (yellow) the active site residue Glu 51 is far from the active site. Upon binding of cyclin A to CDK2 the PSTAIRE helix (blue) rotates 90° and changes its position so that Glu 51 becomes positioned into the active site. (Adapted from P.D. Jeffry et al., *Nature* 376: 313–320, 1995.)



(a)



(b)

Cyclin A - CDK2

Figure 6.19 Schematic diagram of the conformational changes of CDK2 upon cyclin binding. (a) In the inactive form the PSTAIRE helix (red) is oriented such that Glu 51 points away from the ATP binding site (purple) and the T-loop (yellow) blocks the substrate binding site and prevents proteins from binding to CDK2. (b) In the active cyclin-CDK2 complex the PSTAIRE helix is reoriented so that Glu 51 points into the active site and forms a salt bridge to another residue involved in catalysis, Lys 33. The T-loop has drastically changed its conformation and one of its residues, Asp 145, forms ligands to the Mg atom in the active site. The substrate-binding site is now open, proteins can bind and the cyclin-CDK2 complex can phosphorylate Ser/Thr residues and thereby activate the bound proteins.

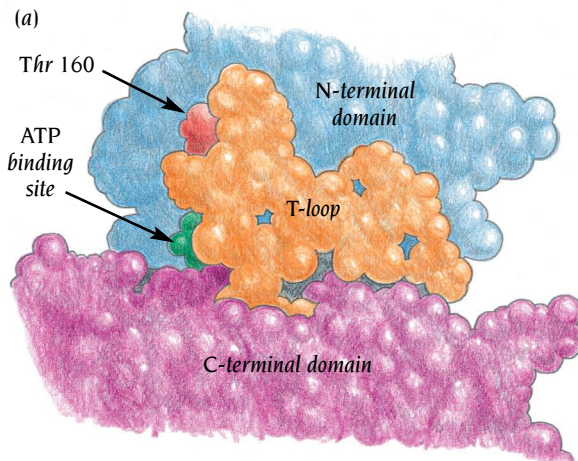
160 is exposed and ready to become phosphorylated, which enhances the catalytic activity. In short, spectacular flexibility of protein structure is essential for regulating the CDK family of enzymes and hence for controlling the cell cycle.

Peptide binding to calmodulin induces a large interdomain movement

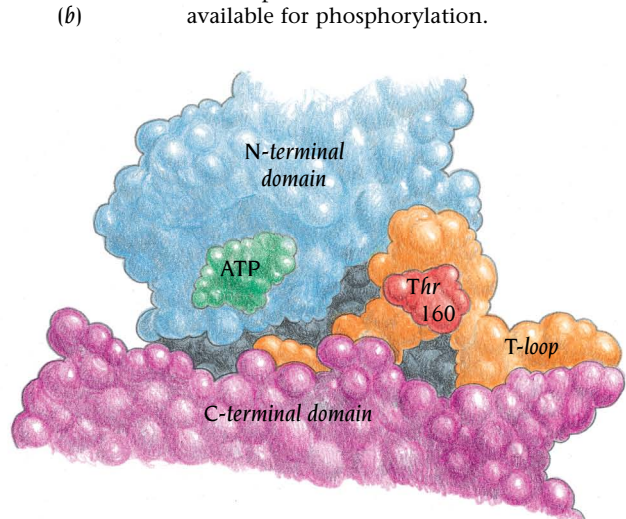
Calmodulin is a ubiquitous calcium-binding protein of 148 amino acid residues that is involved in a range of calcium-dependent signaling pathways. Calmodulin binds to a variety of proteins such as kinases, calcium pumps and proteins involved in motility, thereby regulating their activities. The calmodulin-binding regions of these proteins, comprising about 20 sequentially adjacent residues, vary in their amino acid sequences but they all have a strong propensity to form α helices. Structure determinations of calmodulin alone and of complexes with peptides have shown that peptide binding induces a large conformational change in the calmodulin molecule.

The x-ray structure of free calmodulin was determined by the group of Charles Bugg, University of Alabama. It is a dumbbell-shaped molecule

Figure 6.20 Space-filling diagram illustrating the structural changes of CDK2 upon cyclin binding. (a) The active site is in a cleft between the N-terminal domain (blue) and the C-terminal domain (purple). In the inactive form this site is blocked by the T-loop. (b) In the active cyclin bound form of CDK2 the T-loop has changed its structure, the active site is open and available and Thr 160 is available for phosphorylation.



(a)



(b)

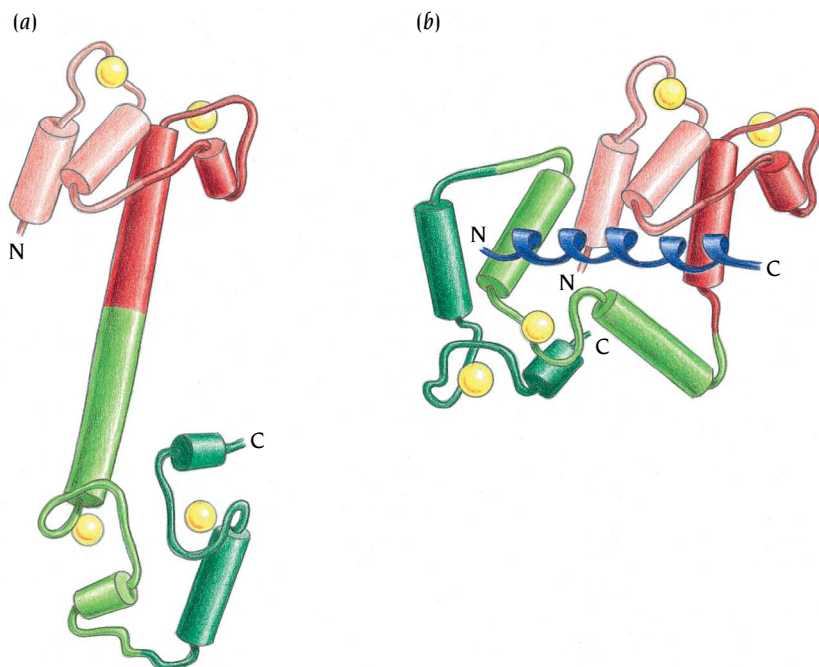


Figure 6.21 Schematic diagram of the conformational changes of calmodulin upon peptide binding. (a) In the free form the calmodulin molecule is dumbbell-shaped comprising two domains (red and green), each having two EF hands with bound calcium (yellow). (b) In the form with bound peptides (blue) the α helix linker has been broken, the two ends of the molecule are close together and they form a compact globular complex. The internal structure of each domain is essentially unchanged. The bound peptide binds as an α helix.

(Figure 6.21a) comprising two domains separated by a long straight α helix, similar in shape to troponin-C described in Chapter 2 (see Figure 2.13c). Each domain comprises two EF hands (see Figure 2.13a), each of which binds a calcium atom. The two domains are clearly separated in space at the two ends of the α helix linker.

The structures of two different complexes of calmodulin with binding peptides have been determined, one by the group of Ad Bax, National Institutes of Health, Bethesda, using NMR and the second by the group of Florante Quiocho, Baylor College of Medicine, Houston, using x-ray crystallography. These two structures are quite similar but the molecular shape of calmodulin in these complexes is very different from that of free calmodulin. The internal structures of the two domains have not changed but the α helix linker has been broken into two helices that are oriented in different directions so that the relative positions of the domains have changed (Figure 6.21b). They are now close together forming a compact molecule of ellipsoidal shape. The bound peptide is in a wide cleft between the two domains and adopts an α -helical conformation.

When calmodulin binds a ligand, only five groups actually change their conformation. They are five consecutive residues in the linker helix, which unwind and turn into a loop region. The α helix continues after this loop but now in an entirely new direction, which positions the second domain close to the first and in a different orientation. This rather small local change in peptide conformation causes one of the largest ligand-induced interdomain motions known in a protein, comparable to the large repositioning of domains during the pH-induced conformational change of hemagglutinin discussed in Chapter 5.

Serpins inhibit serine proteinases with a spring-loaded safety catch mechanism

Infections in the lung elicit an accumulation of activated leucocytes that secrete enzymes involved in removing the damage done by the infection. The most important of these enzymes is neutrophil elastase, which belongs to the serine proteinase family of enzymes described in Chapter 11. The health of the lung depends to a large extent on proper control of the activity of this enzyme, which is achieved by a blood plasma proteinase inhibitor named, misleadingly, α_1 -antitrypsin because it also inhibits other serine proteinases,

among them trypsin. Alpha₁-antitrypsin belongs to a family of serine proteinase inhibitors found in blood plasma that are collectively called **serpins**. Other members of this family are **antithrombin** and **plasminogen activator inhibitor, PAI**, both of which are essential regulators of the blood coagulation cascade of reactions. All serpin molecules are homologous with very similar three-dimensional structures.

Serpins form very tight complexes with their corresponding serine proteinases, thereby inhibiting the latter. A flexible loop region of the serpin binds to the active site of the proteinases. Upon release of the serpin from the complex its polypeptide chain is cleaved by the proteinase in the middle of this loop region and the molecule is subsequently degraded. In addition to the active and cleaved states of the serpins there is also a latent state with an intact polypeptide chain that is functionally inactive and does not bind to the proteinase.

The structures of all three states of the serpins have been determined by x-ray crystallography, the cleaved form of α_1 -antitrypsin by the group of Robert Huber, Max-Planck Institute for Biochemistry, Munich, the latent form of PAI by the group of Elizabeth Goldsmith, University of Texas, Dallas, and the active form of antithrombin by the groups of Wim Hol, University of Washington, Seattle, and of Robin Carrel, Cambridge University. In addition, the group of Robin Carrel has determined the structure of another member of the serpin family, **ovalbumin**, which is present in uncleaved form in egg white. The general folds of these serpin molecules in their different states are the same, but the positions of the flexible loop regions vary in a novel and intriguing way.

The serpin fold comprises a compact body of three antiparallel β sheets, A, B and C, which are partly covered by α helices (Figure 6.22). In the structure of the uncleaved form of ovalbumin, which can be regarded as the canonical structure of the serpins, sheet A has five strands. The flexible loop starts at the end of strand number 5 of β sheet A (β 15 in Figure 6.22), then

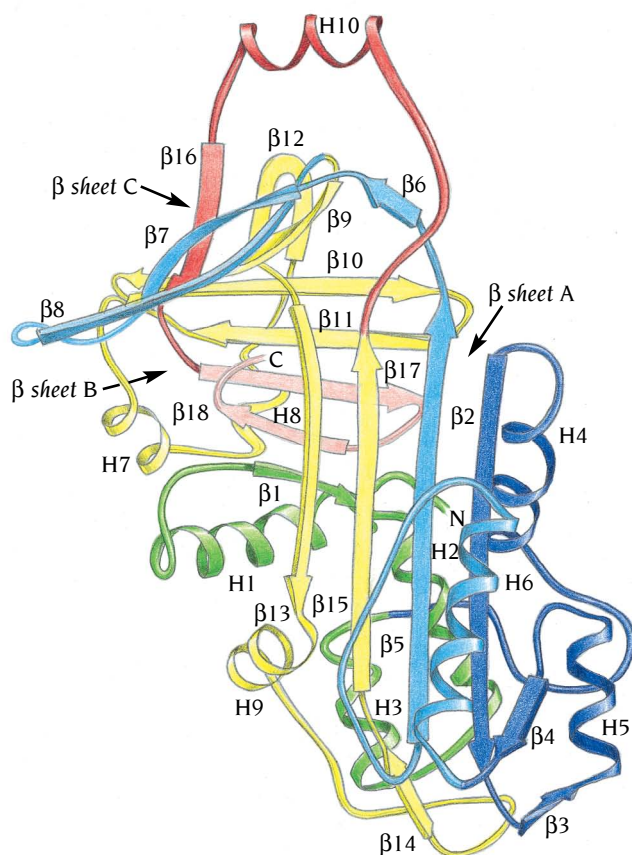


Figure 6.22 Schematic diagram of the structure of ovalbumin which illustrates the serpin fold. The structure is built up of a compact body of three antiparallel β sheets, A, B, and C, surrounded by α helices. The polypeptide chain is colored in sections from the N-terminus to facilitate following the chain tracing in the order green, blue, yellow, red and pink. The red region corresponds to the active site loop in the serpins which in ovalbumin is protruding like a handle out of the main body of the structure. (Adapted from R.W. Carrell et al., *Structure* 2: 257–270, 1994.)

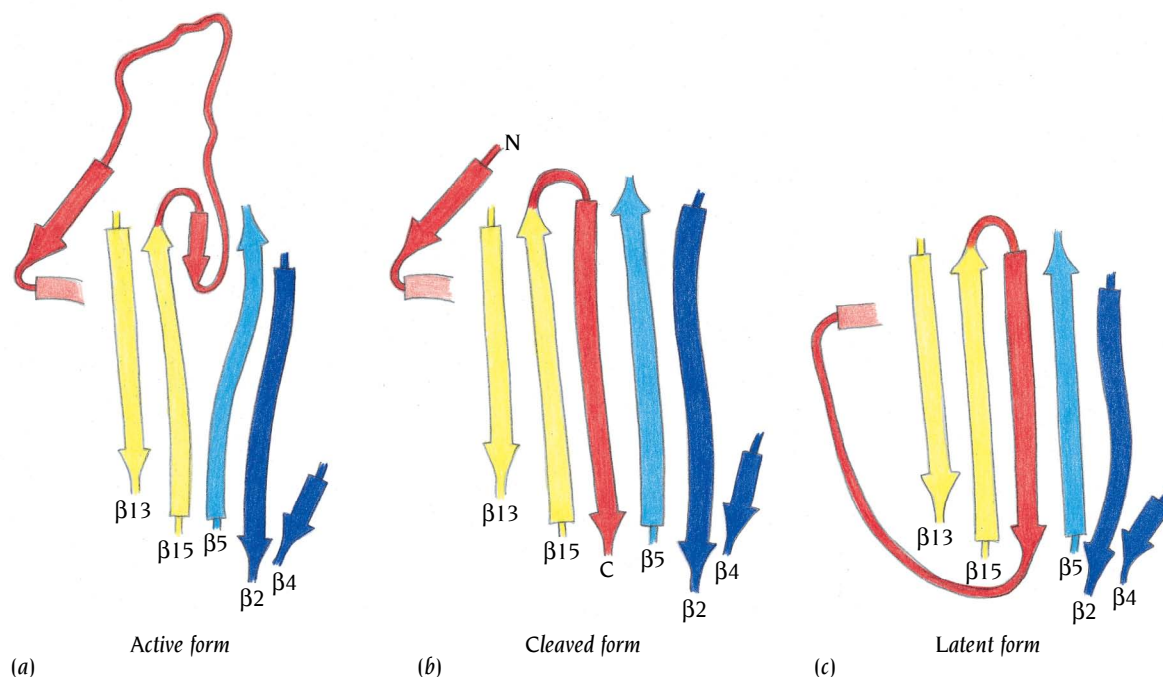


Figure 6.23 Schematic diagram illustrating the active site loop regions (red) in three forms of the serpins. (a) In the active form the loop protrudes from the main part of the molecule poised to interact with the active site of a serine proteinase. The first few residues of the loop form a short β strand inserted between β 5 and β 15 of sheet A. (b) As a result of inhibiting proteases, the serpin molecules are cleaved at the tip of the active site loop region. In the cleaved form the N-terminal part of the loop inserts itself between β strands 5 and 15 and forms a long β strand (red) in the middle of the β sheet. (c) In the most stable form, the latent form, which is inactive, the N-terminal part of the loop forms an inserted β strand as in the cleaved form and the remaining residues form a loop at the other end of the β sheet. (Adapted from R.W. Carrell et al., *Structure* 2: 257–270, 1994.)

forms an α helix outside the top of the molecule followed by an edge strand of β sheet C (β 16) and finally ends at the beginning of one of the strands in β sheet B (β 17). The central α -helical region of the loop, which contains the cleavage site for the serpins, is extended like a handle on the outside of the molecule.

The flexible loop region in the active form of antithrombin (Figure 6.23a) is in the same general position as in ovalbumin but the first few residues form a short sixth β strand in β sheet A inserted between strands β 5 and β 15. Furthermore there is no α helix in the loop which is extended outside the main body of the molecule, ready to be inserted into the active site of thrombin.

In the cleaved form of α_1 -antitrypsin the first half of the loop region up to the cleavage site forms a complete β strand inserted between strands β 5 and β 15 in β sheet A (Figure 6.23b). The other half of the loop region has approximately the same position as in the active form of antithrombin. The two new ends of the polypeptide, which are joined in the active form, are here at opposite ends of the molecule, 70 Å apart. Finally, in the latent form of PAI the additional β strand in β sheet A is present as in the cleaved form of α_1 -antitrypsin, but the rest of the flexible loop region makes a loop on the outside of the molecule and enters the β strand of β sheet B without forming an edge strand of β sheet C (Figure 6.23c).

The conversion of the active form to the latent form involves the conversion of a loop into a long β strand inserted in the middle of a β sheet. To achieve the remarkable structural change of inserting a β strand in the middle of a stable preformed β sheet, adjacent strands in the β sheet must first be separated. This involves breaking many hydrogen bonds as well as changing a number of hydrophobic packing contacts in the interior of the molecule. New hydrogen bonds and new packing contacts must then be made when the extra β strand is inserted. Such major changes in a β structure were quite unforeseen before these serpin structures were determined and have not yet been observed in any other system.

Which of these forms is most stable? Surprisingly, the active form is less stable than the latent form. Conversion from the active to the latent form can occur spontaneously over a period of hours or days *in vitro* and more quickly under mild denaturing conditions. In contrast, recovery of the active form from the latent form requires complete unfolding of the latent form

under strong denaturing conditions, and subsequent refolding. Refolding does, however, produce the less stable active form in preference to the more stable latent form. This is one of the few pieces of direct experimental evidence that the folding process can be kinetically controlled through intermediates that produce a native state that is not the thermodynamically most stable state.

In vivo PAI and antithrombin are stabilized in their active forms by binding to vitronectin and heparin, respectively. These two serpins seem to have evolved what Max Perutz has called “a spring-loaded safety catch” mechanism that makes them revert to their latent, stable, inactive form unless the catch is kept in a loaded position by another molecule. Only when the safety catch is in the loaded position is the flexible loop of these serpins exposed and ready for action; otherwise it snaps back and is buried inside the protein. This remarkable biological control mechanism is achieved by the flexibility that is inherent in protein structures.

Emphysema is often associated with a specific mutation of the serpin antitrypsin. The mutant serpin molecules form aggregates in the liver, causing a deficiency of antitrypsin in the blood plasma and consequently increased proteolytic degradation of elastin fibers in the lung by the enzyme elastase. It has been shown that the formation of aggregates *in vivo* is due to an extremely slow folding process of the mutant antitrypsin leading to accumulation of a folding intermediate that aggregates. This is one example of aggregation of incompletely folded or misfolded molecules that can lead to pathologic consequences or even severe disease. Other examples involve the formation of large aggregates, plaques, of proteins in amyloid structures associated with Alzheimer’s disease and spongiform encephalopathies such as scrapie, BSE “mad cow disease” and Creutzfeld-Jacob disease (see Chapter 14). A better understanding of the folding and misfolding processes might, therefore, open up new approaches to drugs for these diseases.

Effector molecules switch allosteric proteins between R and T states

In 1963 Jaques Monod, Jean-Pierre Changeaux and Francois Jacob published a theory that radically changed our views on the control of protein function and which still influences protein biochemists as well as structural biologists. Their theory of **allosteric control** provided a unifying theme for such diverse concepts as **feedback inhibition** of enzymes, repressor/corepressor binding (see Chapter 8) and **cooperative binding** of ligand by proteins, with oxygen binding to hemoglobin as the prime example. The allosteric theory they proposed has the following main features. Cooperative substrate binding and modification of a protein’s activity by allosteric effector molecules may arise in proteins with two or more preformed structural states that are in equilibrium. Substrates and effector molecules bind at different sites on the protein and therefore need no stereochemical relationship to each other, hence the name allostery (different shapes).

The theory predicts that such proteins are built up of several subunits which are symmetrically arranged and that the two states differ by the arrangements of the subunits and the number of bonds between them. In one state the subunits are constrained by strong bonds that would resist the structural changes needed for substrate binding, and this state would consequently bind substrates weakly; they called it the tense or T state. In the other state, called the R state, these constraints are relaxed.

This **concerted model** assumes furthermore that the symmetry of the molecule is conserved so that the activity of all its subunits is either equally low or equally high, that is, all structural changes are concerted. Subsequently Daniel Koshland, University of California, Berkeley, postulated a **sequential model** in which each subunit is allowed independently to change its tertiary structure on substrate binding. In this model tertiary structural changes in the subunit with bound ligand alter the interactions of this

subunit with its neighbours and this leads sequentially to changes in the latter's reactive sites. Koshland's model was based on his theories of induced fit; ligand binding induces a conformational change that converts an enzyme from an inactive to an active state.

For many years hemoglobin was the only allosteric protein whose stereochemical mechanism was understood in detail. However, more recently detailed structural information has been obtained for both the R and the T states of several enzymes as well as one genetic repressor system, the trp-repressor, described in Chapter 8. We will here examine the structural differences between the R and the T states of a key enzyme in the glycolytic pathway, phosphofructokinase.

X-ray structures explain the allosteric properties of phosphofructokinase

Phosphofructokinase, PFK, is the key regulatory enzyme in glycolysis, the breakdown of glucose to generate ATP which occurs in most cells (Figure 6.24a). The enzyme catalyzes one of the early steps in this pathway, phosphorylation by ATP of fructose-6-phosphate, F6P, to fructose-1,6-bisphosphate (Figure 6.24b). Binding to PFK of one of the substrate molecules, F6P, is highly cooperative whereas binding of the second substrate ATP is

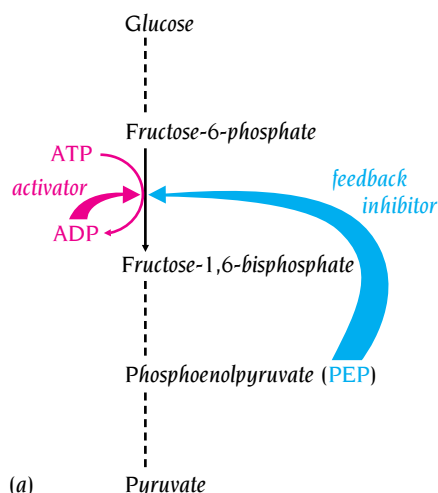
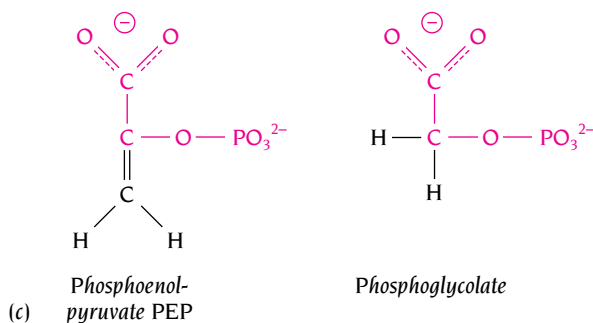
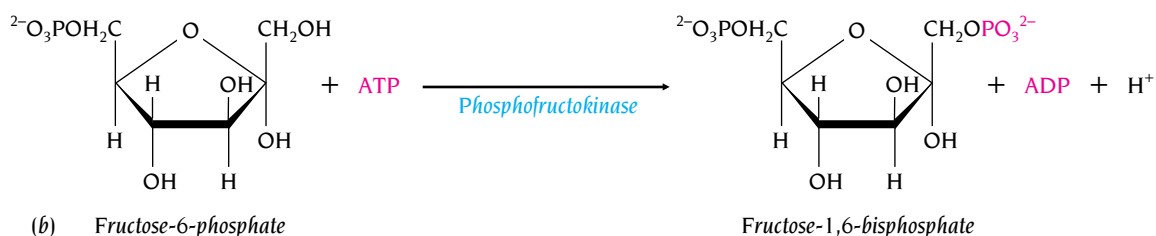


Figure 6.24 The function of the enzyme phosphofructokinase. (a) Phosphofructokinase is a key enzyme in the glycolytic pathway, the breakdown of glucose to pyruvate. One of the end products in this pathway, phosphoenolpyruvate, is an allosteric feedback inhibitor to this enzyme and ADP is an activator. (b) Phosphofructokinase catalyzes the phosphorylation by ATP of fructose-6-phosphate to give fructose-1,6-bisphosphate. (c) Phosphoglycolate, which has a structure similar to phosphoenolpyruvate, is also an inhibitor of the enzyme.



noncooperative. Phosphofructokinase is inhibited by phosphoenolpyruvate, PEP, one of the products of a late step in the glycolytic pathway, and by chemical analogs of PEP, for example 2-phosphoglycolate (Figure 6.24c). By contrast the reaction rate is enhanced by ADP, an allosteric effector molecule. The regulation of PFK by effector molecules is the main way that the glucose degradation by glycolysis is controlled in cells.

Because of the crucial role of this enzyme in one of the most important biochemical pathways in the cell, its allosteric properties have been studied extensively in solution. Interpretation of these studies in terms of the theory of allosteric enzymes led Monod and coworkers to conclude that:

1. The enzyme is made of four identical subunits each having a single binding site for each ligand.
2. The subunits can switch between two distinct conformational states, R and T, which are in equilibrium.
3. The transitions between these states in each tetrameric molecule are concerted, in other words all four subunits of each molecule are in the same state, either R or T.
4. The two states have the same affinity for ATP but differ with respect to their affinity for the substrate F6P, the allosteric effector ADP and the inhibitor PEP. Because of these differences in affinity, ligand binding can shift the equilibrium between the R and T states to favor one or the other state depending on which ligand is bound.

The group of Phil Evans, MRC Laboratory of Molecular Biology, Cambridge, UK, has determined x-ray structures of bacterial PFK both in the R and the T states. These studies have confirmed the above conclusions and given insight into how an allosteric enzyme accomplishes its complex behavior.

Each subunit of the homotetrameric PFK of *Escherichia coli* comprises 320 amino acids arranged in two domains, one large and one smaller, both of which have an α/β structure reminiscent of the Rossman fold (Figure 6.25).

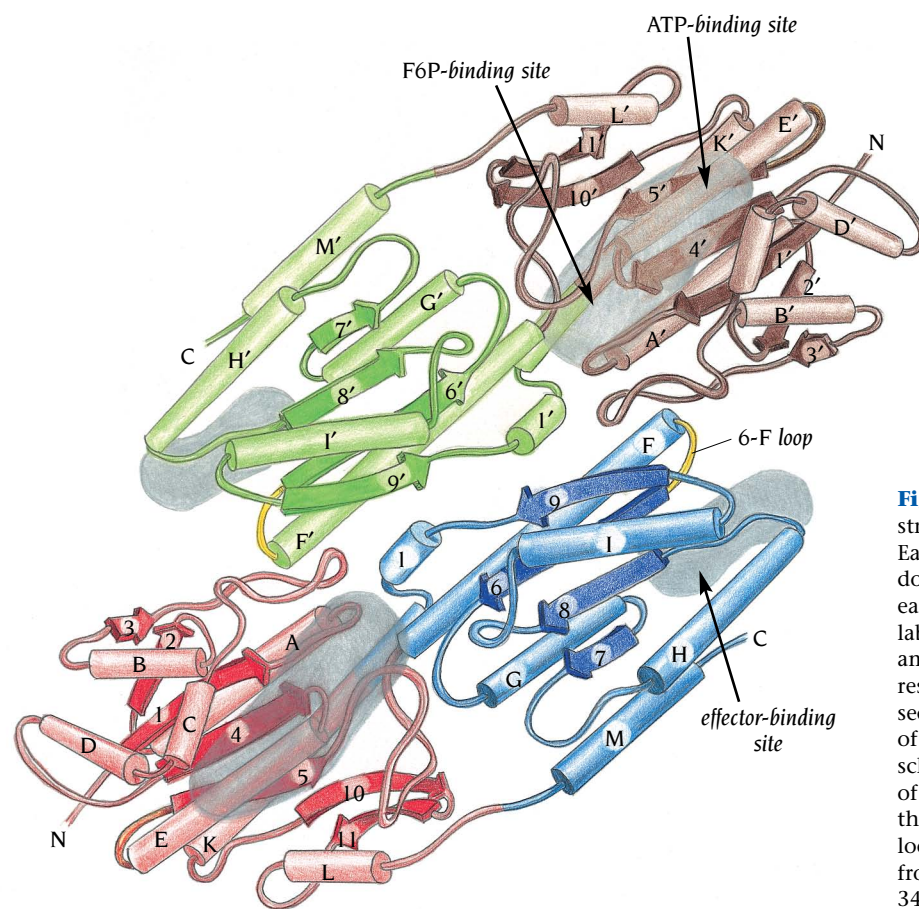


Figure 6.25 Schematic diagram of the structure of one dimer of phosphofructokinase. Each polypeptide chain is folded into two domains (blue and red, and green and brown), each of which has an α/β structure. Helices are labeled A to M and β strands 1 to 11 from the amino terminus of one polypeptide chain, and respectively A' to M' and 1' to 11' for the second polypeptide chain. The binding sites of substrate and effector molecules are schematically marked in gray. The effector site of one subunit is linked to the active site of the other subunit of the dimer through the 6-F loop between helix F and strand 6. (Adapted from T. Schirmer and P.R. Evans, *Nature* 343: 140–145, 1990.)

The subunits are pairwise linked into two dimers (A-B and C-D in Figure 6.26a) with extensive close contacts between the subunits within the dimers. The two dimers are loosely packed against each other into a symmetrical tetramer. The close contacts between the two subunits of the dimer are the same in the R and the T states but the interactions between the dimers are quite different. The orientation of the dimers with respect to each other in the R and T states differs by a rotation of 7°. This difference affects the packing of the dimers against each other and hence the quaternary structure. In the T state the dimers are close together and there are direct hydrogen bonds between two β strands, one from each dimer (Figure 6.26b). In the R state these two β strands are further apart and the gap between them is filled with water molecules that form hydrogen bond bridges (Figure 6.26c). The inclusion or exclusion of water molecules between the dimers is an all-or-none effect that acts like a two-way switch. As we shall see this change in quaternary structure of the tetrameric molecule is intimately linked to differences in tertiary structure of the subunits in the R and T states.

Each of the four subunits contains three binding sites (see Figure 6.25). There is one site for each of the substrate molecules, ATP and F6P, which together form the active site facing a cleft between the two domains. The third binding site of each subunit is the regulator binding site to which both the inhibitor PEP and the allosteric activator ADP can bind; this site is distant from the active site. Evans has studied crystals of the catalytically competent R state of three types: (1) with the subunits complexed with substrates, (2) subunits complexed with ADP, and (3) subunits unliganded. He also studied crystals of the T state in which the subunits are complexed with the inhibitor 2-phosphoglycolate.

The catalytic site of each subunit is in the cleft between the two domains. The large domain binds ATP with the terminal phosphate pointing into the cleft. The main binding site for the second substrate molecule, F6P, is in the smaller domain and the phosphate group of F6P interacts with a neighboring subunit affecting subunit interactions that are crucial for catalytic activity. In the active R state this phosphate group forms hydrogen bonds to an arginine residue, Arg 162, of a small α helix, called the 6-F helix, in the neighboring subunit (Figure 6.27a). By contrast in the T form this helix is unwound and instead forms a loop with Arg 162 pointing away from the F6P molecule (Figure 6.27b). In the T state a negatively charged glutamate side chain, Glu 161, occupies the same position as the positively charged Arg 162 in the R state. This negatively charged glutamate 161 repels the negative charge of the phosphate group of F6P. Consequently, when the R state is transformed to

Figure 6.26 The quaternary structure of phosphofructokinase. (a) The four subunits are pairwise arranged in two dimers A-B (blue) and C-D (red or green). The subunit interactions within the dimers are extensive and tight whereas the two dimers are loosely packed against each other and the packing contacts are different in the R and the T states. The orientation of the dimers with respect to each other in the T (red contours of the C-D dimer) and R (green contours) differs by a rotation of 7°. (b) The dimers are close together in the T state and there are direct hydrogen bonds between two β strands, one from the A-B dimer (blue) and one from the C-D dimer (green). Hydrogen bonds are shown in orange. (c) The dimers are further apart in the R state and there is a gap between the two β strands from the two dimers which is filled by water molecules (red). These water molecules form hydrogen bonds to the C=O and N-H groups of the two β strands.

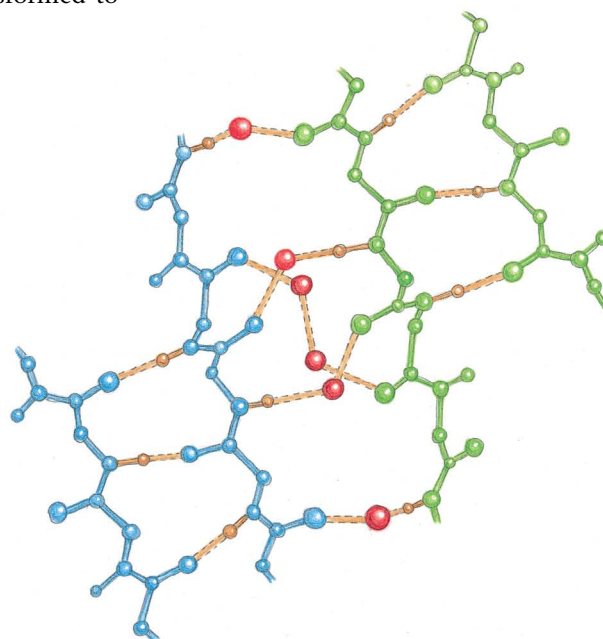
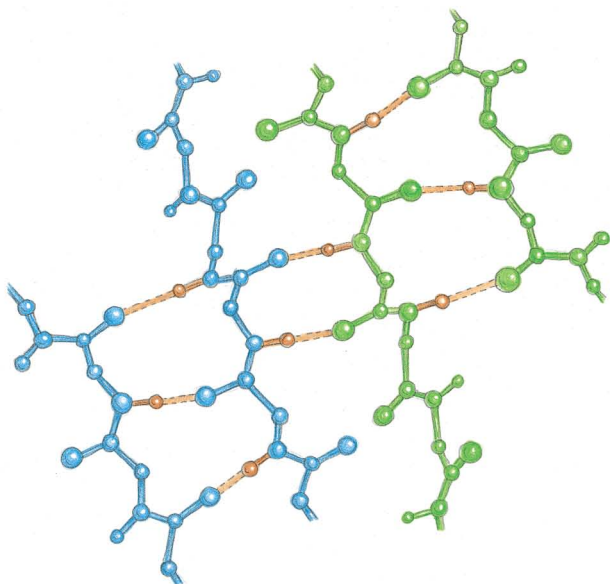
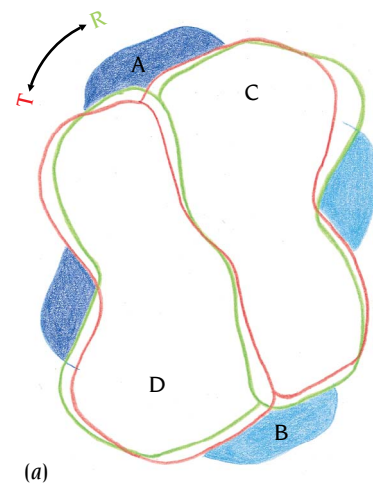
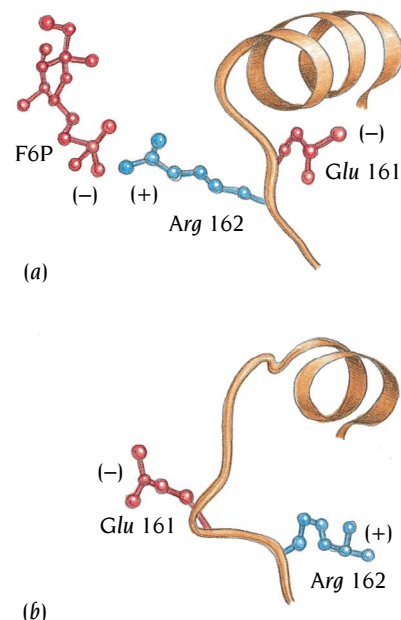


Figure 6.27 Conformational changes in the active site of phosphofructokinase. (a) In the active R state the phosphate group of the substrate fructose-1-phosphate, F6P, (red) forms a salt bridge to an arginine residue, Arg 162, of a small α helix (orange). This salt bridge contributes substantially to promote binding of the substrate to the enzyme. (b) In the inactive T state the helix has been partially unwound and changed its orientation so that Arg 162 points away from the substrate binding site. Instead a negatively charged glutamate residue, Glu 161, points towards the phosphate binding site of the substrate molecule. Repulsive forces between the negative charges of Glu 161 and the phosphate of F6P prevent binding and result in a thousandfold lower affinity for F6P when the enzyme is in the T state compared with the R state.



the T state the change in the F6P binding site results in a thousandfold lower affinity for F6P. In addition, catalytically important residues are properly arranged for catalysis in the R state but not in the T state. Since the binding site of ATP is virtually identical in the R and T states the binding of ATP is not affected by the structural differences between them.

The change in the quaternary structure and the structural change in the 6-F helix as the molecule moves from one state to the other are intimately related. The dimer interactions in the T state are not compatible with the presence of the 6-F helix, which would, if present, clash with the neighbouring dimer. The quaternary structure of the T state requires that the 6-F helix be unwound. Conversely the R state quaternary structure depends on the presence of the 6-F helix.

The basic kinetic properties of this allosteric enzyme are clearly explained by combining Monod's theory and these structural results. The tetrameric enzyme exists in equilibrium between a catalytically active R state and an inactive T state. There is a difference in the tertiary structure of the subunits in these two states, which is closely linked to a difference in the quaternary structure of the molecule. The substrate F6P binds preferentially to the R state, thereby shifting the equilibrium to that state. Since the mechanism is concerted, binding of one F6P to the first subunit provides an additional three subunits in the R state, hence the cooperativity of F6P binding and catalysis. ATP binds to both states, so there is no shift in the equilibrium and hence there is no cooperativity of ATP binding. The inhibitor PEP preferentially binds to the effector binding site of molecules in the T state and as a result the equilibrium is shifted to the inactive state. By contrast the activator ADP preferentially binds to the effector site of molecules in the R state and as a result shifts the equilibrium to the R state with its four available, catalytically competent, active sites per molecule.

Conclusion

The thermodynamic stability of a protein in its native state is small and depends on the differences in entropy and enthalpy between the native state and the unfolded state. From the biological point of view it is important that this free energy difference is small because cells must be able to degrade proteins as well as synthesize them, and the functions of many proteins require structural flexibility.

When a fully extended unfolded polypeptide chain begins to fold, hydrophobic residues tend to be buried in the interior, greatly restricting the number of possible conformations the chain can assume, and therefore allowing proteins to fold in seconds rather than years. Within milliseconds the polypeptide chain achieves the molten globule state, a term used to describe a set of structures that have in common a loosely packed hydrophobic core and some secondary structure. Some proteins have one preferred folding pathway, while others seem to have multiple parallel pathways to the native state. There are certain high energy barriers to folding such as, for

example, the formation of correct disulfide bonds and the isomerization of proline residues. Cells contain enzymes such as protein disulfide isomerases and *cis-trans*-proline isomerases that catalyze these reactions and therefore overcome what otherwise would be an insuperable energy barrier to rapid folding.

The cytoplasm of all cells contains folded proteins and folding polypeptides at high concentrations. Unfolded proteins with exposed hydrophobic patches aggregate easily by non-specific hydrophobic interactions. To circumvent this problem a class of proteins called chaperones have evolved to sequester unfolded polypeptides. The complex structure of one class of multimeric chaperones, the chaperonins GroEL and GroES, has been elucidated and has shed light on how chaperones function. These chaperonins are short cylinders which, because they have hydrophobic residues in the interior, can bind to any unfolded polypeptide that has large exposed hydrophobic patches, regardless of its amino acid sequence. Once the polypeptide chain is shielded inside the chaperonins it is protected from aggregation with other protein molecules. During folding some polypeptide chains go through many cycles of binding and release from the chaperonins.

Even inside crystals all atoms in protein molecules undergo small oscillations. Protein structures determined by x-rays are an average of these breathing structures. In addition to breathing, some proteins undergo large conformational changes in response to ligand binding or to changes in their environment, and these conformational changes are essential for function. The switches that control the successful passage through the eucaryotic cell cycle depend on changes in the conformation of an α helix and a flexible loop region in the cyclin-dependent kinases. In the case of calmodulin, structural changes play a crucial role in calcium signaling pathways. The dumbbell-shaped inactive molecule collapses into a globular structure that binds to regulatory proteins in such pathways. The serpins, a class of specific serine proteinase inhibitors, undergo an extraordinary conformational change: activation of the inactive form involves conversion of a β strand in the middle of a β sheet into a flexible loop and the converse occurs when the active form changes into the latent form.

Many multimeric enzymes and some other multimeric proteins, the classic examples being hemoglobin and phosphofructokinase, PFK, are subject to allosteric control. Allosteric proteins exist in two states, classically known as the R (relaxed) and the T (tense) states. Effector molecules have a high affinity for only one of these states. Therefore when an effector molecule is present it shifts the equilibrium to favor the high affinity state. The binding site for effector molecules is unrelated to and distinct from the active site. In the case of PFK there are two effector molecules, the activator ADP, which shifts the four identical subunits of the enzyme to the enzymatically active R state, and the inhibitor PEP, which shifts the four subunits to the inactive T state. The active site in the R state has a thousandfold higher affinity for the substrate F6P than does the active site of the T state, due to structural differences between the active sites in these two states. In the cell the activity of PFK is controlled through this allosteric mechanism by the relative concentrations of the two effectors. The inhibitor or negative effector is PEP, which is the product of an enzyme downstream of PFK in the glycolytic pathway. As the concentration of PEP increases it inhibits PFK and downregulates the pathway. This is the classic case of feedback inhibition.

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DNA replication, DNA transcription, and the regulation of gene expression all depend upon the recognition of DNA by proteins. By a powerful combination of structural and genetic studies, in recent years we have begun to understand how these functions are achieved. These insights will be described in Chapters 8 to 10, with emphasis on the contribution of structural studies. Before tackling the structures of DNA-binding proteins and the complexes they form with DNA, however, we need to understand the structure of the **double-stranded, base-paired helical** DNA molecule on its own in order to see what possibilities it offers for the recognition of specific sequences by proteins.

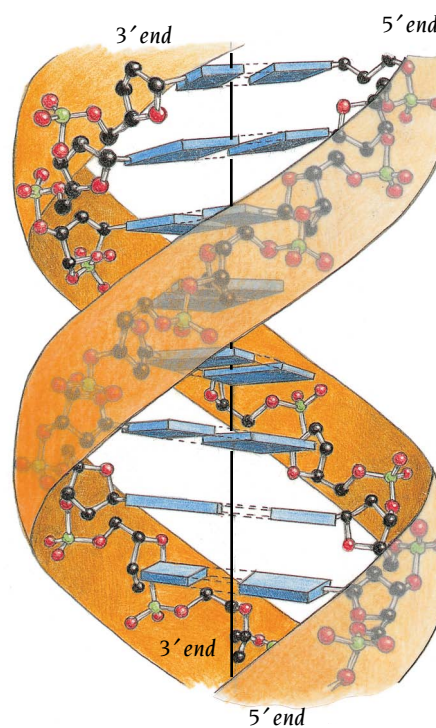
The DNA double helix is different in A- and B-DNA

The DNA molecules in each cell of an organism contain all the genetic information necessary to ensure the normal development and function of that organism. This genetic information is encoded in the precise linear sequence of the nucleotide bases from which the DNA is built. DNA is a linear molecule: while its diameter is only about 20 Å, if stretched out its length can reach many millimeters. This means that concentrated solutions of DNA can be pulled into fibers in which the long thin DNA molecules are oriented with their long axes parallel.

Early diffraction photographs of such DNA fibers taken by Rosalind Franklin and Maurice Wilkins in London and interpreted by James Watson and Francis Crick in Cambridge revealed two types of DNA structures: **A-DNA** and **B-DNA**. The B-DNA form is obtained when DNA is fully hydrated as it is *in vivo*. A-DNA is obtained under dehydrated nonphysiological conditions. Improvements in the methods for the chemical synthesis of DNA have recently made it possible to study crystals of short DNA molecules of any selected sequence. These studies have essentially confirmed the refined fiber diffraction models for A- and B-DNA and in addition have given details of small structural variations for different DNA sequences. Furthermore, a new structural form of DNA, called **Z-DNA**, has been discovered.

Both A-DNA and B-DNA have the familiar shape of a right-handed helical staircase (Figure 7.1). The rails are two antiparallel phosphate-sugar chains, and the rungs are purine-pyrimidine base pairs, which are hydrogen bonded to each other. In A-DNA there are an average of 10.9 base pairs per turn of the helix, which corresponds to an average helical-twist angle of 33.1° ($10.9 \times 33.1 = 360^\circ$) from one base pair to the next. The spacing along the helix axis from one base pair to the next is 2.9 Å. In B-DNA these values are

Figure 7.1 Schematic drawing of B-DNA. Each atom of the sugar-phosphate backbones of the double helix is represented as connected circles within ribbons. The two sugar-phosphate backbones are highlighted by orange ribbons. The base pairs that are connected to the backbone are represented as blue planks. Notice that in B-DNA the central axis of this double helix goes through the middle of the base pairs and that the base pairs are perpendicular to the axis.



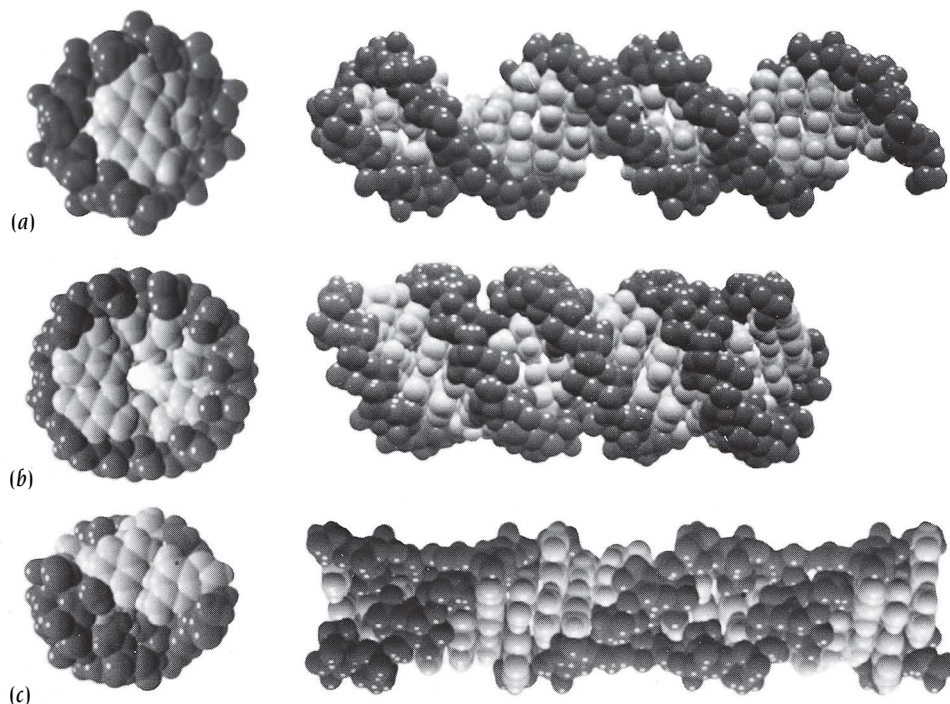


Figure 7.2 Three helical forms of DNA, each containing 22 nucleotide pairs, shown in both side and top views. The sugar-phosphate backbone is dark; the paired nucleotide bases are light. (a) B-DNA, which is the most common form in cells. (b) A-DNA, which is obtained under dehydrated nonphysiological conditions. Notice the hole along the helical axis in this form. (c) Z-DNA, which can be formed by certain DNA sequences under special circumstances. (Courtesy of Richard Feldmann.)

10.0 base pairs and 35.9° and $3.4 \text{ \AA}$, respectively. There are, however, considerable variations in individual twist angles from the average values, and these variations are larger in A-DNA than in B-DNA. These variations are sequence dependent, and in B-DNA they might be important for the specificity of interactions with proteins.

The DNA helix has major and minor grooves

The sugar-phosphate backbones are bulky and form ridges on the edges of the helix, with grooves in between within which the bases are exposed (Figure 7.2). These grooves are of two different widths, reflecting the asymmetrical attachment of the base pairs to the sugar rings of the backbone. Whereas in a regular helix the distance between the attachment points for the rungs would be the same at the front and the back of each step (Figure 7.3), in the DNA molecule each base-pair “rung” is effectively wider at one edge than at the other (Figure 7.4) so that the helical molecule has one narrower groove, known as the **minor groove**, and one wider groove, known as the **major groove** (Figure 7.5).

In B-DNA because the helical axis runs through the center of each base pair and the base pairs are stacked nearly perpendicular to the helical axis (see Figures 7.1 and 7.5), the major and minor grooves are of similar depths.

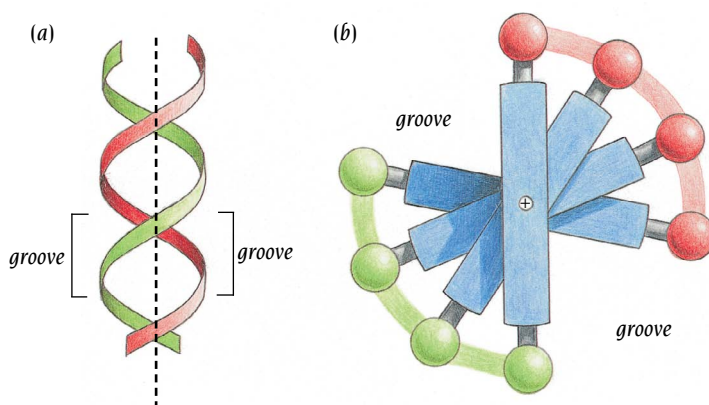
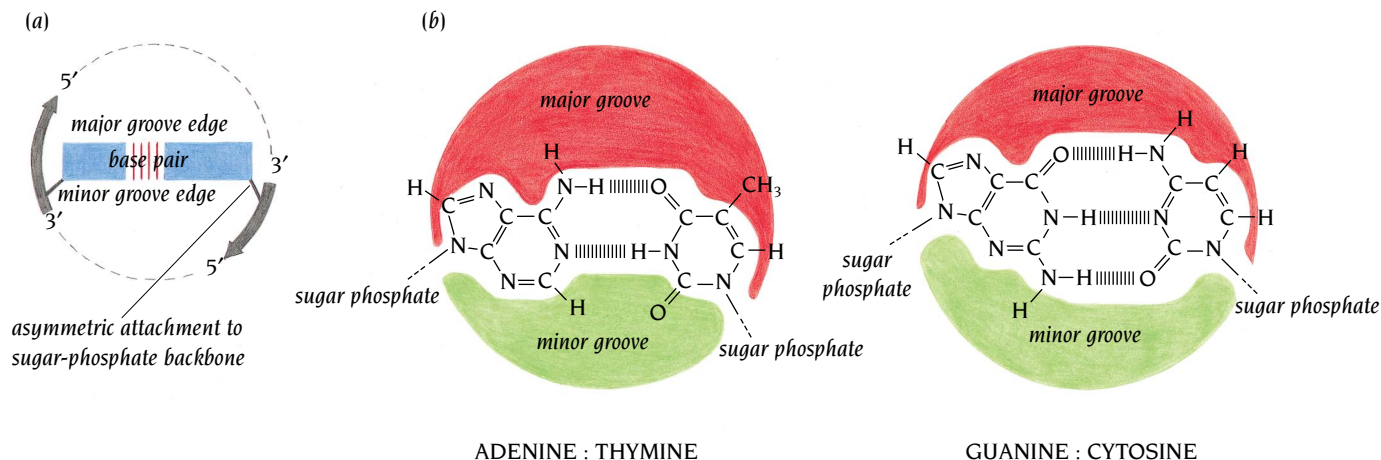


Figure 7.3 Schematic diagram illustrating that there are two similar grooves in a helical staircase. Four rungs are viewed from the top of the staircase in (b).



In A-DNA, on the other hand, the helical axis is shifted from the center of the bases into the major groove, bypassing the bases, and the base pairs are not perpendicular to this axis but are tilted between 13° and 19° . This arrangement makes the major groove very deep, extending from the surface all the way past the central axis and part of the way out toward the opposite side, while the minor groove is shallow, scarcely more than a helical depression spiraling around the outside of the cylinder (see Figure 7.5).

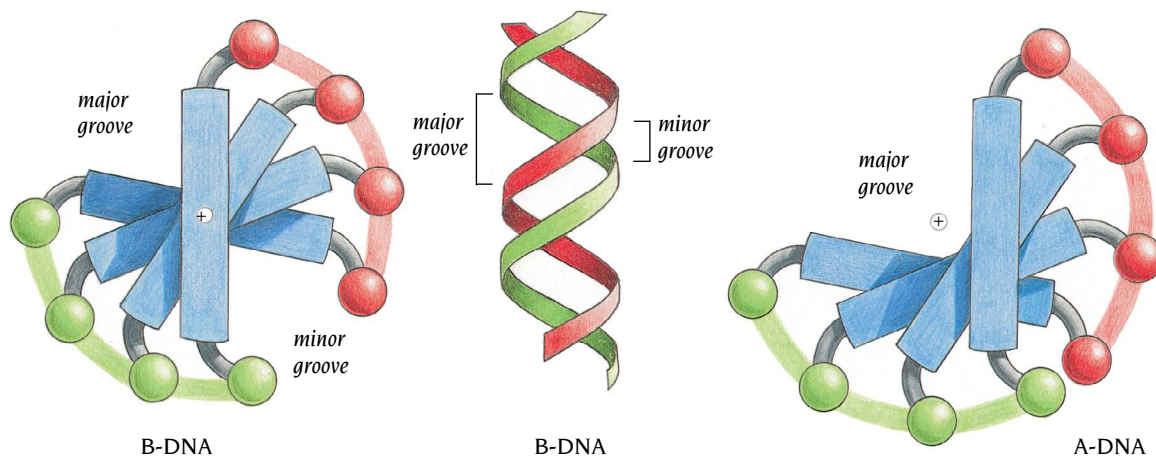
The edges of the base pairs form the floors of the two grooves. The edge of a base pair furthest from its attachment points to the sugar-phosphate backbones is the major groove edge; the one closest is the minor groove edge (see Figure 7.4a). These edges are accessible from the outside and form the basis for the sequence-specific recognition of DNA by proteins.

Z-DNA forms a zigzag pattern

Z-DNA has a quite different structure (see Figure 7.2c). The helix is left-handed, and the sugar-phosphate backbone follows a zigzag path. The structure has been found for sequences with alternating G and C bases, such as CGCG and CGCGCG. Each cytosine has its sugar attached to the base in such a way that the pyrimidine ring swings away from the minor groove. This is the normal conformation for all four bases in A- and B-DNA. However, each guanine in Z-DNA has its sugar ring rotated 180° so that it bends inward toward the minor groove. The sugar-phosphate backbone of alternating C and G bases with their different conformations make a zigzag pattern around the helix. The shape of this helix is thin and elongated. It has a deep but quite narrow minor groove, whereas the major groove is pushed to the surface so that it is no longer a groove at all.

Figure 7.4 The edges of the base pairs in DNA that are in the major groove are wider than those in the minor groove, due to the asymmetric attachment of the base pairs to the sugar-phosphate backbone (a). These edges contain different hydrogen bond donors and acceptors for potentially specific interactions with proteins (b).

Figure 7.5 Schematic diagram illustrating the major and minor grooves in A- and B-DNA. The sugar-phosphate backbone is represented by connected circles in color and the base pairs as blue planks. Four base pairs are shown from the top of the helix to highlight how the grooves are formed due to the asymmetric connections. The position of the helix axis is marked by a cross.



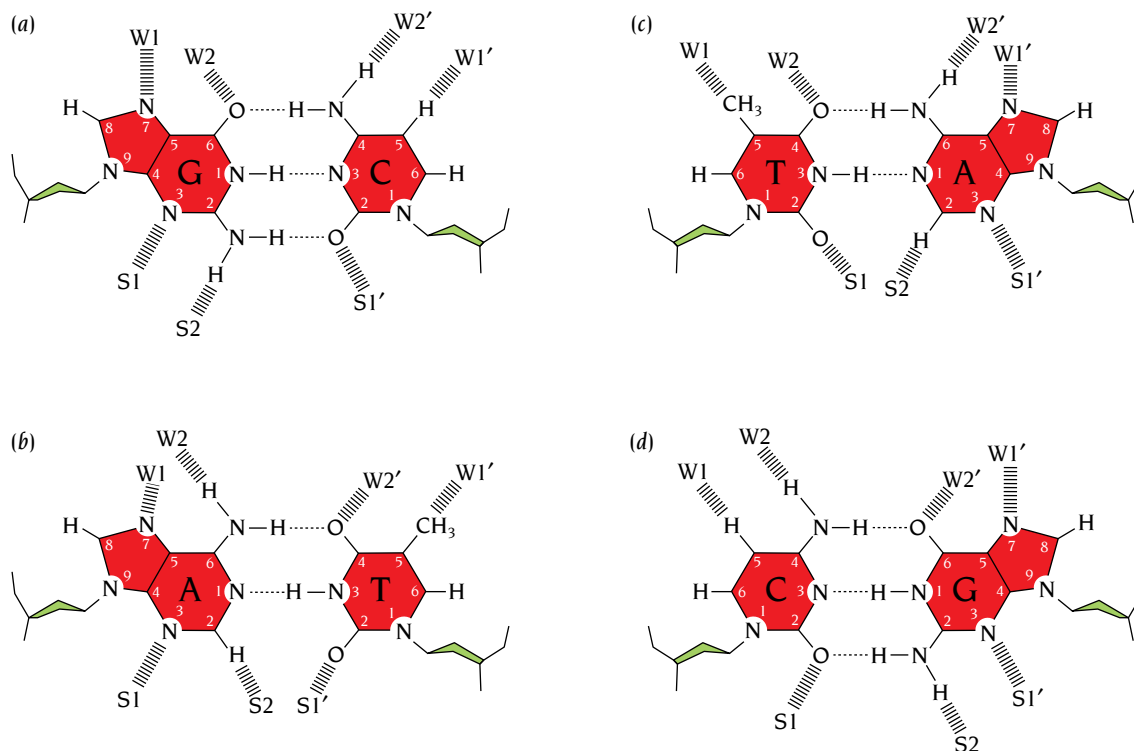


Figure 7.6 The edges of the base pairs contain nitrogen and oxygen atoms that can make hydrogen bonds to protein side chains. An H atom in cytosine (C) and a methyl group in thymine (T) form additional sequence-specific recognition sites in DNA. W1, W2, W2', and W1' are the recognition sites at the edges of the base pairs in the major groove (W for wide) and S1, S2, and S1' are those in the minor groove (S for small). The recognition sites are shown for all four base pairs: GC (a), AT (b), TA (c), and CG (d).

B-DNA is the preferred conformation *in vivo*

The specific protein–DNA interactions described in this book are all with DNA in its regular B-form, or, in some cases with distorted B-DNA. In biological systems DNA appears not to adopt the A conformation, although double-stranded RNA does preferentially adopt this conformation *in vivo*. Whether or not Z-DNA occurs in nature is a matter of controversy. However, the formation of A-DNA and Z-DNA *in vitro* does illustrate the large structural changes that DNA can be forced to undergo.

Specific base sequences can be recognized in B-DNA

Let us now look at how specific sequences might in principle be recognized in B-DNA. The only regions where the bases are available for interaction are at the floor of the grooves. These are paved with nitrogen and oxygen atoms that can make hydrogen bonds with the side chains of a protein. The methyl group of thymine and the corresponding hydrogen in cytosine provide additional discriminatory recognition groups (see Figure 7.4). These sites, which are represented in Figure 7.6 and schematically illustrated in Figure 7.7 in the form of a color code, form patterns that are different for the four possible Watson/Crick base pairs. If, for instance, we compare a GC and an AT base-pair reading from the purine to the pyrimidine ring, we can read the

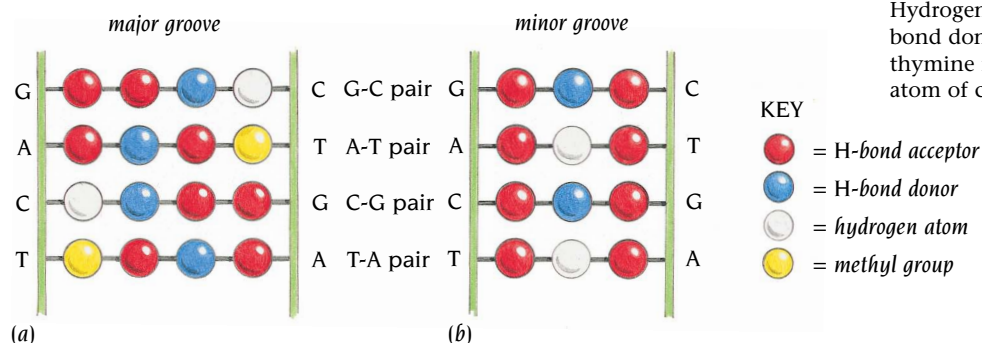


Figure 7.7 Color codes for the recognition patterns at the edges of the base pairs in the major (a) and minor (b) grooves of B-DNA. Hydrogen-bond acceptors are red; hydrogen-bond donors are blue. The methyl group of thymine is yellow, while the corresponding H atom of cytosine is white.

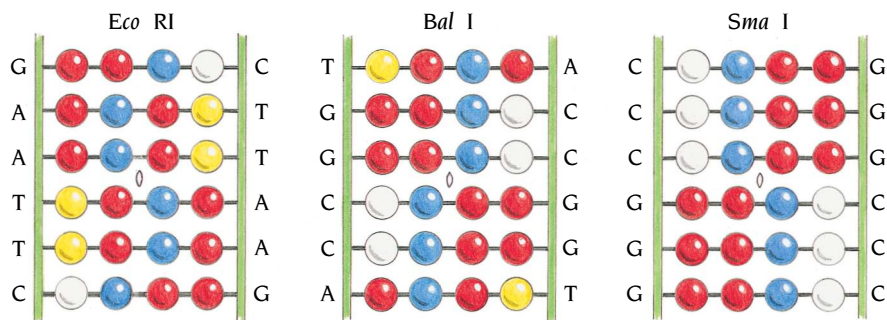


Figure 7.8 Sequence-specific recognition sites in the major groove of DNA for three restriction enzymes—Eco RI, Bal I, and Sma I. The DNA sequences that are recognized by these enzymes are represented by the color code defined in Figure 7.7.

recognition pattern in the following way. For the GC pair, the major groove exposes a hydrogen-bond acceptor, nitrogen 7 of guanine (G N7; W1), another acceptor, G O6 (W2), a hydrogen-bond donor, C NH4 (W2'), and, finally, a hydrogen atom at C5 (W1') (see Figure 7.6a). For the AT pair, the corresponding groups are acceptor A N7, donor A NH6, acceptor T O4, and a methyl group at T5 (see Figure 7.6b).

These patterns of potential hydrogen-bond acceptors and donors are clearly quite different for the different base pairs in the major groove, and they could easily be recognized and distinguished by a protein molecule. This is not the case in the minor groove. If we look at the same two base pairs, we see that for the GC pair we have an acceptor G N3 (S1), a donor G NH2 (S2), and an acceptor C O2 (S1'). For the AT pair we find the following pattern: an acceptor A N3, a hydrogen atom at A2, and an acceptor T O2. In the minor groove there is only one distinguishing feature between these two base pairs: a hydrogen-bond donor group in GC compared with a neutral hydrogen atom in AT. We can represent these recognition patterns by a color code as shown in Figure 7.7. Hydrogen-bond donors and acceptors of the base pairs have different colors as well as the H and CH₃ groups of cytosine and thymine, respectively. In this way, each of the four different base pairs is represented by a unique color code, comprising four positions in the major groove and three in the minor groove. Clearly, the major groove is a much better candidate for sequence-specific recognition than the minor groove for two reasons. First, the major groove is wider than the minor, and the bases are thus more accessible to a protein molecule. Second, the pattern of possible hydrogen bonds from the edges of the base pairs to a protein are more specific and discriminatory in the major groove than in the minor (see Figure 7.7).

Only a rather limited number of base pairs is needed to provide unique and discriminatory recognition sites in the major groove. This is illustrated in Figure 7.8, which gives the color codes for the hexanucleotide recognition sites of three different restriction enzymes—Eco RI, Bal I, and Sma I. It is clear that these patterns are quite different, and each can be uniquely recognized by specific protein–DNA interactions.

Bacteriophage repressor proteins provide excellent examples of sequence-specific interactions between the side chains of a protein and bases lining the floor of the major groove of B-DNA. As we shall see, to fit the protein's recognition module into this groove it has to be made even wider; in other words, the B-DNA has to be distorted.

Conclusion

B-DNA is the conformation of DNA *in vivo*. It has a wide major groove and a narrow minor groove. The edges of the base pairs form the bottom of the grooves, where nitrogen and oxygen atoms are available to make hydrogen bonds with the side chains of a protein. Such hydrogen bonds form the basis for sequence-specific recognition of DNA.

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Structure, Function, and Engineering

Part 2

DNA Recognition in Prokaryotes by Helix-Turn-Helix Motifs

8

Proteins that regulate transcription of DNA recognize specific DNA sequences through discrete DNA-binding domains within their polypeptide chains. These domains are in general relatively small, less than 100 amino acid residues. Many prokaryotic DNA-binding domains contain a **helix-turn-helix motif** that recognizes and binds specific regulatory regions of DNA. The two α helices have the same orientation relative to each other, and they are connected by a loop region of similar structure in all DNA-binding helix-turn-helix motifs. In this chapter we will discuss the functional properties of the helix-turn-helix motif and the way this motif is integrated into structurally different DNA-binding domains of prokaryotic repressors and activators. In the following two chapters we discuss the wider range of DNA binding motifs in eukaryotic DNA-binding proteins.

The mechanism of action of bacterial and bacteriophage repressors and activators is in principle very simple. Repressors bind tightly to the DNA at the promoter of a structural gene, preventing the RNA polymerase from gaining access and hence blocking the initiation of transcription. Activators, on the other hand, work by binding next to the promoter and helping the polymerase to bind to the adjacent promoter, thereby increasing the rate of transcription of the gene. However, as we will see, the subtle regulation of these binding interactions can be quite complex.

Some of the most thoroughly studied prokaryotic regulator proteins belong to bacteriophage lambda and related phages. These phages produce two regulator proteins, namely, **repressor** and **Cro**. Repressor and Cro were given their names at an early date and subsequent studies have shown that, paradoxically, lambda Cro is a repressor protein, whereas lambda repressor has both repressor and activator functions; the names, however, have been kept for historical reasons. Repressor and Cro proteins operate a switch between two states of lambda phage replication. Both proteins contain the helix-turn-helix motif. Even though this motif was first observed in a different protein, CAP—which activates operons involved in sugar breakdown in *Escherichia coli*—we will focus our discussion on the two bacteriophage proteins since they are amongst the best-understood DNA regulatory proteins.

A molecular mechanism for gene control

Certain strains of *Escherichia coli* can be stimulated by irradiation with a moderate dose of ultraviolet (UV) light to stop normal growth and start producing bacteriophages that eventually lyse the bacterium. Bacteria of these so-called lysogenic strains carry the DNA of the phage integrated into their own

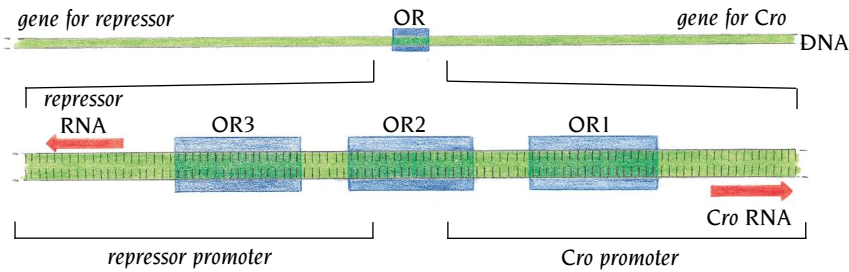


Figure 8.1 A region of DNA in the related bacteriophages lambda, 434, and P22 that controls the switch for synthesis of new phage particles. Two structural genes are involved in this switch; one coding for a repressor protein and one coding for the Cro protein. Between these genes there is an operator region (OR) that contains three protein binding sites—OR1, OR2, and OR3.

chromosomes, where it is dormant during normal cell growth; the phage DNA is replicated as an integral part of the bacterial chromosome, but the phage genes are not expressed. The phage in this state is called a temperate phage. Ultraviolet light switches on the phage genes which then produce new phages and the cell eventually dies. We are now able to explain in part the molecular mechanism of this genetic on-off switch from results obtained by a powerful combination of genetic and x-ray structural studies.

Repressor and Cro proteins operate a procaryotic genetic switch region

Three related species of temperate bacteriophages have been studied, lambda, 434, and P22. A relatively small region of the phage genome contains all the genetic components of the on-off switch (Figure 8.1). In each of the three species of phage this region comprises two structural genes coding for the two regulator proteins, Cro and repressor, that operate the switch; and the “right” operator region (OR) on which they act.

The two genes are transcribed in opposite directions from their two promoters, which occupy opposite ends of the operator region (Figure 8.1). When RNA polymerase is bound to the left-hand promoter, repressor is switched on, *Cro* and the lytic genes are repressed (Figure 8.2a), and the cell survives as a lysogenic strain. When the polymerase is bound to the right-hand promoter, *Cro* is switched on, along with the early lytic genes that lie to the right of *Cro*, and cell lysis results (Figure 8.2b).

The lysis-lysogeny decision depends upon which of the two promoters in the operator region is able to bind polymerase, and that, in turn, depends upon the binding of the Cro and repressor proteins to three binding sites—OR1, OR2, and OR3—in OR. These binding sites are situated in the middle of the operator in such a way that OR1 and OR2 overlap the promoter

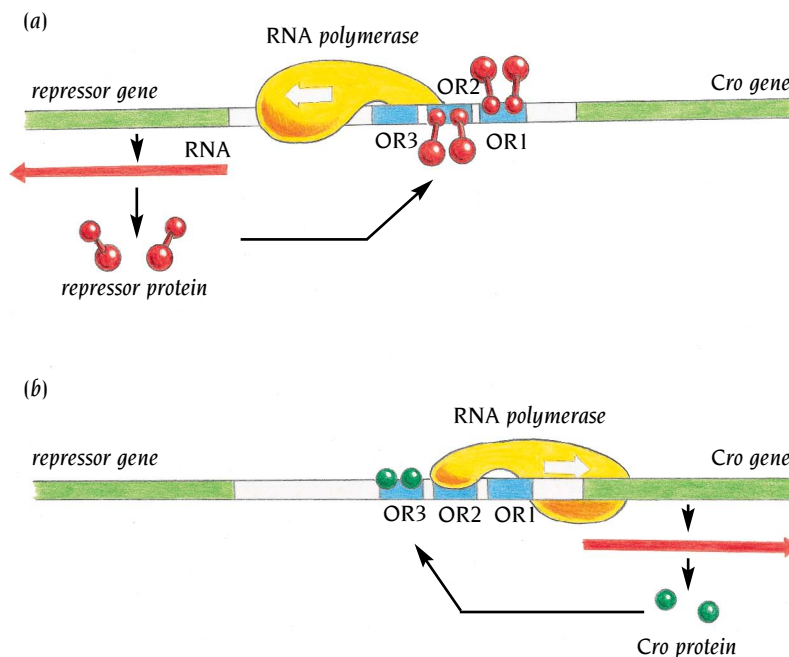


Figure 8.2 (a) The situation in the lysogenic bacterium. Repressor (red) and RNA polymerase (yellow) bound to the switch region in a lysogenic strain of *E. coli*. The repressor binds to OR1 and OR2, thereby turning off synthesis of *Cro*. The repressor also works as an activator for its own synthesis by facilitating RNA-polymerase binding to the repressor promoter through its binding to OR2. (b) The situation in the lytic phase. Synthesis of the *Cro* protein turns off synthesis of the repressor, since *Cro* binds to OR3 and blocks RNA-polymerase binding to the repressor promoter. Transcription of phage genes to the right can now occur.

of the *Cro* gene and OR2 and OR3 overlap the promoter for the repressor gene (see Figure 8.1). Both Cro and repressor bind as dimers to all three sites, but Cro binds with the highest affinity to OR3, and when it is bound, it blocks the access of RNA polymerase to the promoter for the repressor gene; conversely, repressor binds with the highest affinity to OR1, and when it is bound, it blocks the access of the polymerase to the *Cro* promoter.

Thus, repressor establishes the lysogenic state by binding to OR1 and switching off the lytic genes to the right of *Cro*. At the same time, it ensures the maintenance of this state by binding to OR2, where instead of blocking the access of the polymerase to its own promoter, it helps the polymerase to bind, thus producing more repressor. In the lysogen, repressor dimers are more or less continuously bound to OR1 and OR2 because once a molecule of repressor is bound at its high-affinity site at OR1, it helps a second molecule to bind at OR2 through a cooperative interaction between the two molecules. Thus repressor acts both to repress Cro synthesis and to activate its own synthesis.

Cro, by contrast, acts purely as a repressor. When it is bound to its high-affinity site at OR3, it prevents repressor synthesis by obstructing the access of polymerase to the left-hand promoter. In the absence of repressor, RNA polymerase can bind to the *Cro* promoter, and Cro can be synthesized along with the early phage genes to its right.

Thus, for a lysogen to switch over to the production of phage particles, repressor must be released from OR1 for long enough to allow Cro to be synthesized and bind to OR3. This is brought about by ultraviolet radiation, which activates a bacterial protease, Rec A, that cleaves the repressor molecules into two so that they can no longer form stable dimers. The protease activity of Rec A is activated when polymers of Rec A bind to single-stranded DNA formed as a result of DNA damage by UV light. The cleaved repressor fragments have a much lower affinity for OR1 and OR2 than the intact repressor molecule, and after a while the *Cro* promoter site becomes free to bind RNA polymerase, allowing transcription of the *Cro* gene along with the early genes for viral replication. The newly formed Cro protein molecules bind to OR3, blocking the access of RNA polymerase to the repressor promoter, and synthesis of more repressor molecules is prevented (Figure 8.2b). The switch has been flipped, and the machinery of the cell is converted to the synthesis of phage particles.

The x-ray structure of the complete lambda Cro protein is known

How do repressor and Cro recognize the specific operator regions and achieve this subtle differential binding to the switch regions? The sequences of OR1, OR2, and OR3 in lambda are similar but not identical (Table 8.1). They are

Table 8.1 The nucleotide sequences of the three protein-binding regions OR1, OR2, and OR3 of the operator of bacteriophage lambda

		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	
OR1	5'	T	A	T	C	A	C	C	G	C	C	A	G	T	G	G	T	A	3'
	3'	A	T	A	G	T	G	G	C	G	G	T	C	A	C	C	A	T	5'
OR2	5'	T	A	A	C	A	C	C	G	T	G	C	G	T	G	T	T	G	3'
	3'	A	T	T	G	T	G	G	C	A	C	G	C	A	C	A	A	C	5'
OR3	5'	T	A	T	C	A	C	C	G	C	A	A	G	G	G	A	T	A	3'
	3'	A	T	A	G	T	G	G	C	G	T	T	C	C	C	T	A	T	5'

Palindromic base pairs that are most frequent at the two ends are green, and the pseudo-twofold symmetry axis is indicated by a red dot.

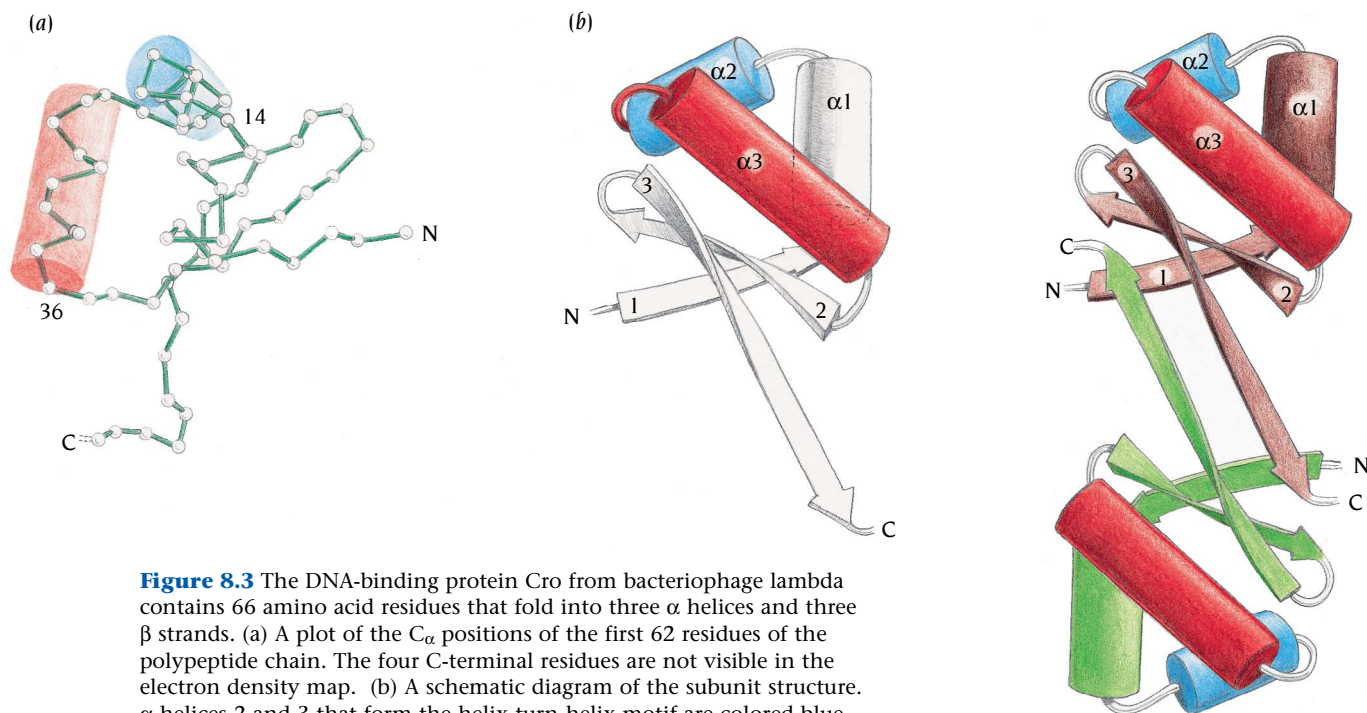


Figure 8.3 The DNA-binding protein Cro from bacteriophage lambda contains 66 amino acid residues that fold into three α helices and three β strands. (a) A plot of the C_{α} positions of the first 62 residues of the polypeptide chain. The four C-terminal residues are not visible in the electron density map. (b) A schematic diagram of the subunit structure. α helices 2 and 3 that form the helix-turn-helix motif are colored blue and red, respectively. The view is different from that in (a). [(a) Adapted from W.F. Anderson et al., *Nature* 290: 754–758, 1981. (b) Adapted from D. Ohlendorf et al., *J. Mol. Biol.* 169: 757–769, 1983.]

also partly **palindromic**, especially at their ends. The palindromic nature of these sequences is important because it means that the halves of each binding site are related by an approximate twofold symmetry axis. Both Cro and repressor proteins are dimers, and the palindromic parts of OR1, OR2, and OR3 provide them with two almost but not quite identical recognition sites, one for each dimer subunit.

A first glimpse into the structural details of this system was obtained in 1981 when the group of Brian Matthews at Eugene, Oregon, determined the crystal structure of the Cro protein of phage lambda at 2.8 Å resolution. Cro is a small protein that forms stable dimers in solution. Each subunit is a single polypeptide chain of 66 amino acid residues with a very simple structure. It folds into three α helices and three strands of antiparallel β sheet and belongs therefore to the $\alpha + \beta$ class of structures, with the three α helices in a loop region between β strands 1 and 2 (Figure 8.3).

However, the α helices are not packed against each other in the usual way as described in Chapter 3. Instead, α helices 2 and 3, residues 15–36, form a unique helix-turn-helix arrangement that in 1981 had only been observed once, in a different bacterial DNA-binding protein, the catabolite gene-activating protein CAP.

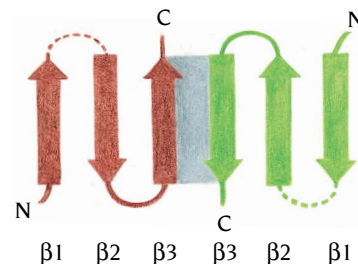
Dimerization of pairs of Cro monomers depends primarily on interactions between β strand 3 from each subunit (Figure 8.4). These strands, which are at the carboxy end of the chains, are aligned in an antiparallel fashion and hydrogen bonded to each other so that the three-stranded β sheets of the monomers form a six-stranded antiparallel β sheet in the dimer (Figure 8.5).

The x-ray structure of the DNA-binding domain of the lambda repressor is known

Superficially, the lambda repressor protein is very different from lambda Cro. The polypeptide chain is much larger, 236 amino acids, and is composed of two domains that can be released as separate fragments by mild proteolysis. In repressor the domain responsible for dimerization is separate from the

Figure 8.4 Cro molecules from bacteriophage lambda form dimers both in solution and in the crystal structure. The main dimer interactions are between β strands 3 from each subunit. In the diagram one subunit is green and the other is brown. Alpha helices 2 and 3, the helix-turn-helix motifs, are colored blue and red, respectively, in both subunits. (Adapted from D. Ohlendorf et al., *J. Mol. Biol.* 169: 757–769, 1983.)

Figure 8.5 The three β strands of each subunit in lambda Cro are aligned in the dimer so that a six-stranded antiparallel β sheet is formed as shown in this topology diagram. The β strands are colored as in Figure 8.4.



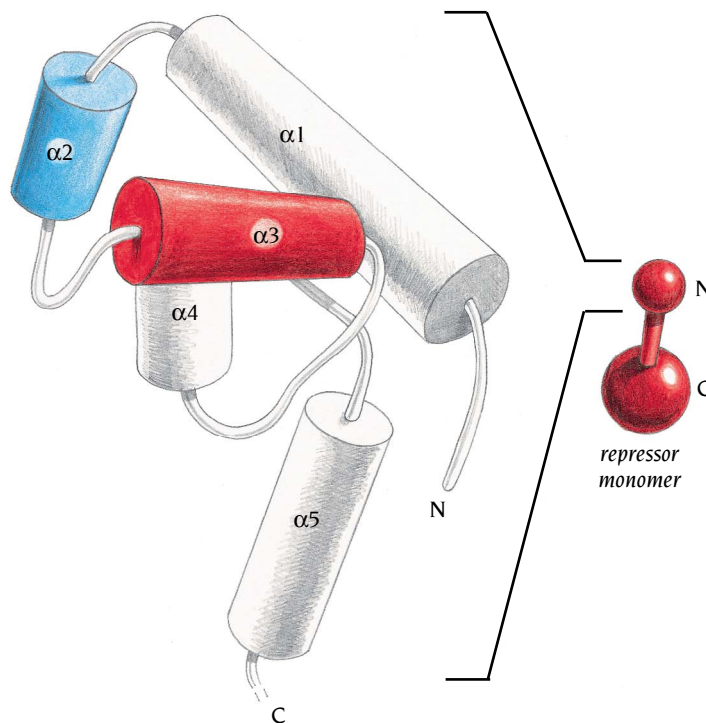


Figure 8.6 The N-terminal domain of lambda repressor, which binds DNA, contains 92 amino acid residues folded into five α helices. Two of these, $\alpha 2$ (blue) and $\alpha 3$ (red) form a helix-turn-helix motif with a very similar structure to that of lambda Cro shown in Figure 8.4. The complete repressor monomer contains in addition a larger C-terminal domain. (Adapted from C. Pabo and M. Lewis, *Nature* 298: 443–447, 1982.)

domain with DNA-binding functions; the C-terminal domains form the strong subunit interactions that hold the dimer together, while the N-terminal domains of 92 residues bind specifically to operator DNA. Although the N-terminal domains can form dimers on their own, they do so only weakly, and for this reason they also bind the operator more weakly than does the intact repressor molecule. This feature is crucial for the switch from lysogeny to the lytic cycle: the proteolytic cleavage that is activated by UV light separates the C-terminal domain from the N-terminal domain and this is how the affinity of repressor for OR1 is reduced and the switch from lysogeny to lysis initiated.

The x-ray structure of the N-terminal DNA-binding domain of the lambda repressor was determined to 3.2 Å resolution in 1982 by Carl Pabo at Harvard University and revealed a structure with striking similarities to that of Cro, although the β strands in Cro are replaced by α helices in repressor.

The polypeptide chain of the 92 N-terminal residues is folded into five α helices connected by loop regions (Figure 8.6). Again the helices are not packed against each other in the usual way for α -helical structures. Instead, α helices 2 and 3, residues 33–52, form a helix-turn-helix motif with a very similar structure to that found in Cro.

In spite of the absence of the C-terminal domains, the DNA-binding domains of lambda repressor form dimers in the crystals, as a result of interactions between the C-terminal helix number 5 of the two subunits that are somewhat analogous to the interactions of the C-terminal β strand 3 in the Cro protein (Figure 8.7). The two helices pack against each other in the normal way with an inclination of 20° between the helical axes. The structure of the C-terminal domain, which is responsible for the main subunit interactions in the intact repressor, remains unknown.

Both lambda Cro and repressor proteins have a specific DNA-binding motif

The specific arrangement of two α helices joined by a loop region in lambda Cro and repressor, as well as in CAP, constitute the helix-turn-helix DNA-binding motif (Figure 8.8), which also occurs in some eucaryotic transcription factors as discussed in Chapter 9. The orientation of the two helices and

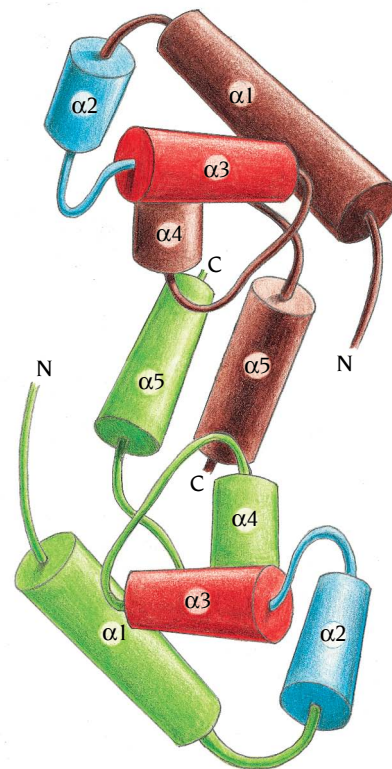


Figure 8.7 The N-terminal domains of lambda repressor form dimers, in spite of the absence of the C-terminal domains that are mainly responsible for dimer formation in the intact repressor. The dimers are formed by interactions between α helix 5 from each subunit. The different subunits are colored green and brown, except the helix-turn-helix motif, which is colored blue and red as in Figure 8.4. (Adapted from C. Pabo and M. Lewis, *Nature* 298: 443–447, 1982.)

the conformation of the loop regions are very similar in these three molecules, and this particular helix-turn-helix structure is unique to DNA-binding proteins.

Since these protein structures were determined in the absence of DNA, there was at that time no experimental evidence that these α helices were involved in DNA binding. However, Brian Matthews in 1981 realized from the structure of Cro that the subunit interactions in the dimer have an important consequence. They cause the second α helix (red in Figure 8.8) in the helix-turn-helix motifs of the two subunits to be at opposite ends of the elongated dimeric molecule, separated by a distance of 34 Å (see Figure 8.4). This distance corresponds almost exactly to one turn of a B-DNA double helix. As a result, if the second α helix of one subunit binds into the major groove of B-DNA, the corresponding α helix of the other subunit can also bind into the major groove one turn further along the DNA molecule. This α helix (the second in the helix-turn-helix motif, colored red in the illustrations) was, therefore, called the **recognition α helix**, and Matthews proceeded to build a model of a possible Cro–DNA complex.

Model building predicts Cro–DNA interactions

Matthews was able to show, by model building on a graphics display, that the two recognition helices of the Cro dimer indeed fitted very well into the major groove of a piece of regular B-DNA as seen in Figure 8.9. The orientation

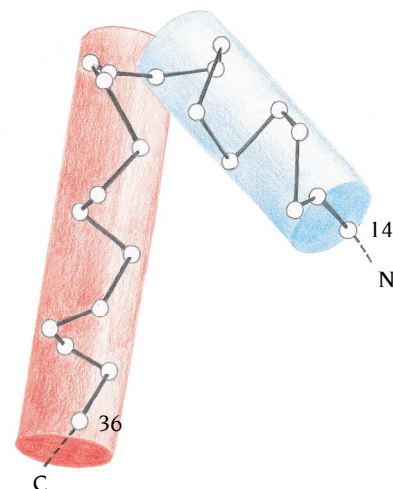


Figure 8.8 The DNA-binding helix-turn-helix motif in lambda Cro. C_{α} positions of the amino acids in this motif have been projected onto a plane and the two helices outlined. The second helix (red) is called the recognition helix because it is involved in sequence-specific recognition of DNA.

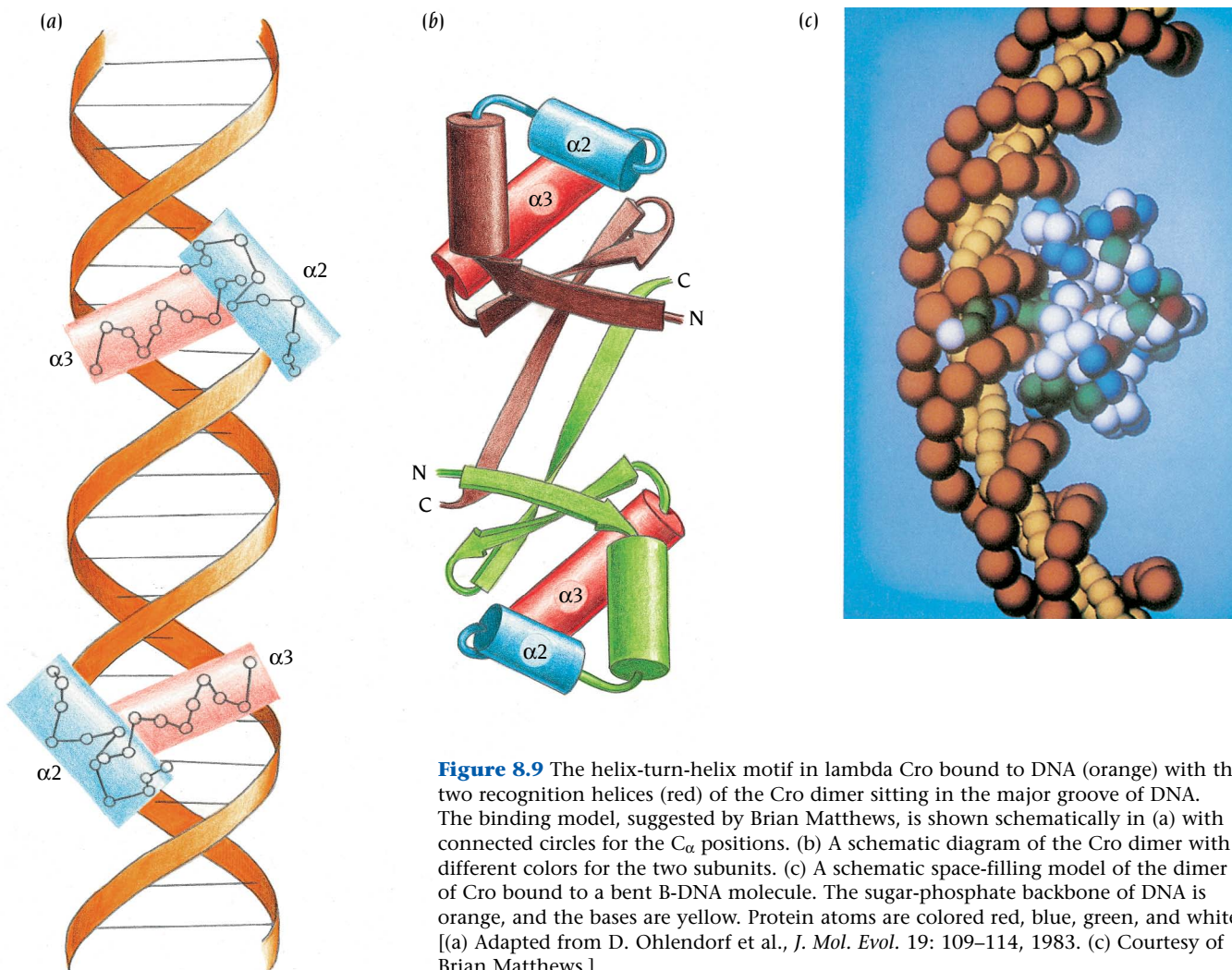


Figure 8.9 The helix-turn-helix motif in lambda Cro bound to DNA (orange) with the two recognition helices (red) of the Cro dimer sitting in the major groove of DNA. The binding model, suggested by Brian Matthews, is shown schematically in (a) with connected circles for the C_{α} positions. (b) A schematic diagram of the Cro dimer with different colors for the two subunits. (c) A schematic space-filling model of the dimer of Cro bound to a bent B-DNA molecule. The sugar-phosphate backbone of DNA is orange, and the bases are yellow. Protein atoms are colored red, blue, green, and white. [(a) Adapted from D. Ohlendorf et al., *J. Mol. Evol.* 19: 109–114, 1983. (c) Courtesy of Brian Matthews.]

of the two recognition α helices follows the orientation of the groove at both sites. Amino acid residues from these α helices can make contact with the edge of the base pairs in the major groove. These amino acids were thus assumed to be involved in recognizing specific operator regions.

Approximately 10 base pairs are required to make one turn in B-DNA. The centers of the palindromic sequences in the DNA-binding regions of the operator are also separated by about 10 base pairs (see Table 8.1). Thus if one of the recognition α helices binds to one of the palindromic DNA sequences, the second recognition α helix of the protein dimer is poised to bind to the second palindromic DNA sequence.

Both the relative orientation of the α helices and the distance between them are determined by the way the dimer is formed through interactions between the subunits. Three features of the Cro structure therefore were claimed to be important for specific binding to operator DNA: (1) the presence of a helix-turn-helix motif that provides a recognition α helix, which binds in the major groove of B-DNA; (2) the specific amino acid sequence of this α helix, which recognizes different operator regions; and (3) the subunit interactions that provide the correct distance and relative orientation between the two recognition α helices of the dimer, thereby increasing the affinity between Cro and operator DNA.

This model of Cro binding to DNA was arrived at by intuition and clever model building. Its validity was considerably strengthened when the same features were subsequently found in the DNA-binding domains of the lambda-repressor molecule. The helix-turn-helix motif with a recognition helix is present in the repressor, and moreover the repressor DNA-binding domains dimerize in the crystals in such a way that the recognition helices are separated by 34 Å as in Cro.

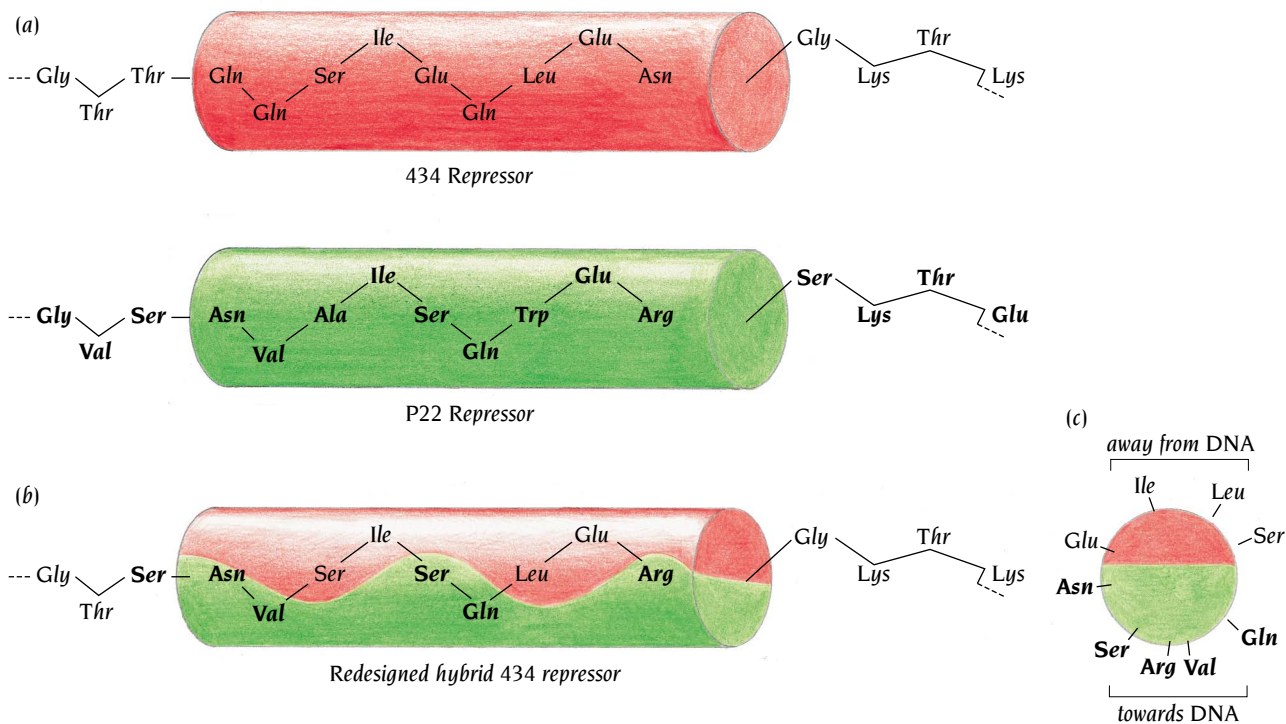
Genetic studies agree with the structural model

The presence of this common helix-turn-helix motif poised for DNA binding in lambda Cro and repressor provided considerable stimulus for further genetic and structural studies of these and other procaryotic DNA-binding proteins. All the results essentially supported the proposed mode of binding between these regulator proteins and DNA.

The most informative genetic experiments were made by the group of Mark Ptashne at Harvard University. In one experiment they redesigned the repressor from phage 434, replacing its recognition α helix with the corresponding α helix from Cro of phage 434 (a difference of five amino acids). The redesigned repressor thereby acquired the differential binding properties of the 434 Cro protein. In another experiment the group changed selected amino acid residues in the recognition α helix of the 434 repressor to those that occur in the repressor of phage P22 (Figure 8.10). The amino acids selected to be changed were those that in the structural model face the DNA and therefore presumably interact with it.

The consequences of these changes were impressive. The parental 434 repressor had no affinity for the P22 operator *in vivo*, but the redesigned 434 repressor controlled only P22 operators and not 434 operator regions *in vivo*. Purified redesigned 434 repressor bound specifically to P22 operator DNA *in vitro* and showed the same hierarchy of affinities for the three regions OR1, OR2, and OR3 as native P22 repressor.

These genetic experiments clearly demonstrated that the proposed structural model for the binding of these proteins to the phage operators was essentially correct. The second α helix in the helix-turn-helix motif is involved in recognizing operator sites as well as in the differential selection of operators by P22 Cro and repressor proteins. However, a note of caution is needed: many other early models of DNA-protein interactions proved to be misleading, if not wrong. Modeling techniques are more sophisticated today but are still not infallible and are certainly not replacements for experimental determinations of structure.



The x-ray structure of DNA complexes with 434 Cro and repressor revealed novel features of protein–DNA interactions

The general features of the model for DNA binding were confirmed experimentally in 1987 when Stephen Harrison's group at Harvard University determined the structure of a complex of DNA and the DNA-binding domain of the 434 repressor to 3.2 Å resolution. However, it also became evident, both from the structure of this complex and from further site-directed mutagenesis studies, that the selective recognition of the different operator regions by the 434 repressor depends mostly on other factors than the amino acid residues of the recognition helix. The complexity of the fine tuning of DNA regulation has been clearly demonstrated by Harrison's subsequent studies of complexes between different operator DNA regions and both 434 Cro and the DNA-binding domain of 434 repressor.

For purely practical reasons, the complexes that Harrison first studied contained the N-terminal DNA-binding domain of the repressor from phage 434, which comprises 69 amino acids, complexed with a 14 base-pair piece of synthetic DNA ("14mer") having a completely palindromic sequence. In other words, the DNA in the complex had a strict twofold symmetry analogous to the twofold symmetry of the dimeric repressor molecule. This synthetic DNA thus contains identical halves, each of which, as we shall see, binds one subunit of Cro or one repressor fragment. The three 14 base-pair operator regions that the 434 repressor recognizes in the phage right-hand operator (OR) are not perfectly palindromic, however (Table 8.2). The 14mer is closest in sequence to a site (OL2) in a second operator region of the phage, the "left" operator (OL), as Table 8.2 shows. The only difference is an inversion of base pair 7 from A–T to T–A, and experiments had shown that this inversion did not alter the affinity for the DNA of either intact repressor or the N-terminal DNA-binding fragment.

The crystals of the complex with 14mer diffracted to only medium resolution, however, and by systematic variations of the length of the DNA fragment and its sequences at the ends, Harrison and coworkers were later able to find a piece of DNA that gave crystals that diffracted to high resolution, both with 434 Cro and the DNA-binding domain of 434 repressor. This DNA fragment contains 20 nucleotides in each chain, and the sequence of its middle region is identical to OR1 (see Table 8.2). The 5' ends contain one non-paired nucleotide that is involved in packing the fragments in the crystal.

Figure 8.10 The proposed DNA-binding surface of the recognition helix of bacteriophage 434 repressor was redesigned genetically to that of P22 repressor by changing six amino acid residues. The amino acid sequences of the recognition helices of the wild-type repressors are shown in (a) and that of the redesigned repressor in (b) and (c) viewed from the side and along the helix, respectively. The redesigned 434 repressor acquired all the DNA-binding properties of the P22 repressor. The six amino acid residues that were changed in the 434 repressor are shown in boldface in (b) and (c). (Adapted from R. Wharton and M. Ptashne, *Nature* 316: 601–605, 1985.)

Table 8.2 The six operator regions (OR1–OR3 and OL1–OL3) in bacteriophage 434

		1	2	3	4	5	6	7	8	9	10	11	12	13	14	
OR1	5'	A	C	A	A	G	A	A	A	G	T	T	T	G	T	3'
	3'	T	G	T	T	C	T	T	T	C	A	A	A	C	A	5'
OR2	5'	A	C	A	A	G	A	T	A	G	T	T	T	G	T	3'
	3'	T	G	T	T	C	T	A	T	C	A	A	A	C	A	5'
OR3	5'	A	C	A	A	G	A	A	A	A	A	C	T	G	T	3'
	3'	T	G	T	T	C	T	T	T	T	T	G	A	C	A	5'
OL1	5'	A	C	A	A	G	G	A	A	G	A	T	T	G	T	3'
	3'	T	G	T	T	C	C	T	T	C	T	A	A	C	A	5'
OL2	5'	A	C	A	A	T	A	A	A	T	A	T	T	G	T	3'
	3'	T	G	T	T	A	T	T	T	A	T	A	A	C	A	5'
OL3	5'	A	C	A	A	T	G	G	A	G	T	T	T	G	T	3'
	3'	T	G	T	T	A	C	C	T	C	A	A	A	C	A	5'
Synthetic DNA	5'	A	C	A	A	T	A	T	A	T	A	T	T	G	T	3'
	3'	T	G	T	T	A	T	A	T	A	T	A	A	C	A	5'
		14'	13'	12'	11'	10'	9'	8'	7'	6'	5'	4'	3'	2'	1'	

Each operator region is 14 base pairs long. The palindromic base pairs of these regions are marked in green. Crystal structures have been determined of complexes between both 434 Cro and the repressor fragment with synthetic DNA fragments—one 14 base pairs long (a 14mer), which is completely palindromic, and one 20 base pairs long (a 20mer), which contains the sequence of OR1 in its middle region.

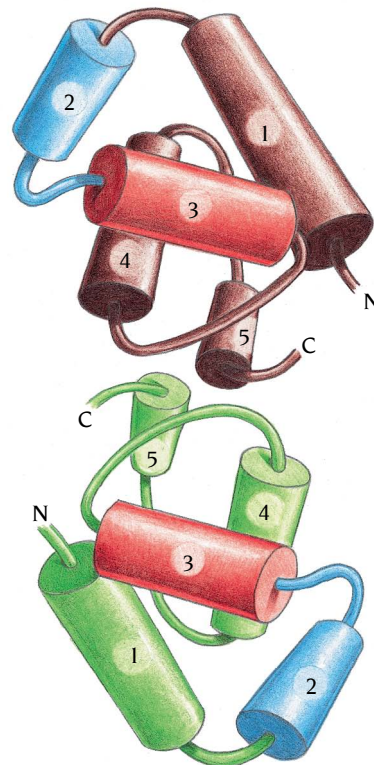
By comparing the crystal structures of these complexes with a further complex of the 434 repressor DNA-binding domain and a synthetic DNA containing the operator region OR3, Harrison has been able to resolve at least in part the structural basis for the differential binding affinity of 434 Cro and repressor to the different 434 operator regions.

The structures of 434 Cro and the 434 repressor DNA-binding domain are very similar

The 434 Cro molecule contains 71 amino acid residues that show 48% sequence identity to the 69 residues that form the N-terminal DNA-binding domain of 434 repressor. It is not surprising, therefore, that their three-dimensional structures are very similar (Figure 8.11). The main difference lies in two extra amino acids at the N-terminus of the Cro molecule. These are not involved in the function of Cro. By choosing the 434 Cro and repressor molecules for his studies, Harrison eliminated the possibility that any gross structural difference of these two molecules can account for their different DNA-binding properties.

The DNA-binding domain of 434 repressor also has significant sequence homology (26% identity) with the corresponding part of the lambda repressor and, consequently, a related three-dimensional structure (compare Figures 8.7 and 8.11). Like its lambda counterpart, the subunit structure of the DNA-binding domain of 434 repressor, as well as that of 434 Cro, contains a cluster of four α helices, with helices 2 and 3 forming the helix-turn-helix motif. The two helix-turn-helix motifs are at either end of the dimer and contribute the main protein–DNA interactions, while protein–protein interactions at the C-terminal part of the chains hold the two subunits together in the complexes. Both 434 Cro and repressor fragments are monomers in solution even at high protein concentrations, whereas they form dimers when they are bound to DNA. (It should be noted once again, however, that in the intact repressor the main dimerization interactions are believed to be formed by the C-terminal domain, which is not present in the crystal complex.)

Figure 8.11 The DNA-binding domain of 434 repressor. It is a dimer in its complexes with DNA fragments. Each subunit (green and brown) folds into a bundle of four α helices (1–4) that have a structure similar to the corresponding region of the lambda repressor (see Figure 8.7) including the helix-turn-helix motif (blue and red). A fifth α helix (5) is involved in the subunit interactions, details of which are different from those of the lambda repressor fragment. The structure of the 434 Cro dimer is very similar to the 434 repressor shown here.



The proteins impose precise distortions on the B-DNA in the complexes

The real significance of Harrison's work with these protein–DNA complexes lies not in the structure of the protein domains but rather in the details of the structures of the bound DNA and in the protein–DNA interactions. The DNA in all the complexes is in the B-form but with significant distortions. Examination of the local twist between base pairs showed that the DNA was overwound (larger twist) at its center and underwound at the ends. The helical axis was also somewhat bent toward the recognition helices at the ends of the protein dimer. These distortions narrow the minor groove at the center and widen it at the ends, as is shown in Figure 8.12, which compares the observed structure of DNA complexed with 434 repressor fragments to that of regular B-DNA. Since the same distortions have been observed in complexes of the 434 repressor fragment with various DNA sequences (the 14mer, the OR1-containing 20mer, an OR2-containing 20mer, and others) and in more than one crystal packing, it is reasonable to conclude that the conformational characteristics of the DNA fragment in these complexes result directly from the protein–DNA interactions.

In complexes with Cro, the overall bend and twist of the DNA are similar to those in the repressor complexes, but there is a significant difference in the local structure of two of the nucleotides in each half-site. Binding of 434 Cro or repressor fragment thus imposes a distinct local structure (Figure 8.13), as a result of differences in both the identity and conformations of various amino acid residues that interact with the DNA. The DNA conformational details are significant for the relative affinities of Cro and repressor for various sites, as we describe in a later section.

Sequence-specific protein–DNA interactions recognize operator regions

The protein–DNA interactions have been analyzed in detail at high resolution in the complex between the 434 repressor fragment and the OR1 containing 20mer DNA. A pseudo-twofold symmetry axis relates the halves of this complex. The symmetry is not exact since the nucleotide sequence of the DNA is slightly different in each half (see Table 8.2). However, the interactions between one protein subunit and one half of the DNA are very similar to those between the second subunit and the other half of the DNA since most of the bases that interact with the protein are identical in both halves. Details of the interaction are very similar to those in the complex with the palindromic synthetic 14mer of DNA shown in Figures 8.14 and 8.15. The base pairs at one end of the DNA, 1–14', 2–13', etc. are called base pairs 1, 2, etc.

The protein dimer binds so that the recognition α helices at opposite ends of the protein molecule are in the major groove of the DNA as predicted, where they interact with base pairs at the end of the DNA molecule. Since these binding sites are separated by one turn of the DNA helix, it follows that at the center of the DNA molecule the narrow groove faces the protein

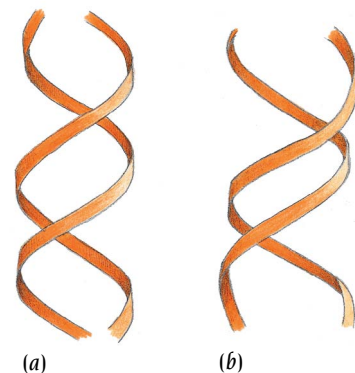


Figure 8.12 (above) The changes of DNA structure from regular B-DNA (a) to a distorted version (b) when 434 Cro and repressor fragment bind to operator regions. The distortions essentially involve bending of DNA and overwinding of the middle regions. The diagram shows the sugar-phosphate backbones of DNA as orange ribbons viewed in the narrow groove in the middle region of the operator.

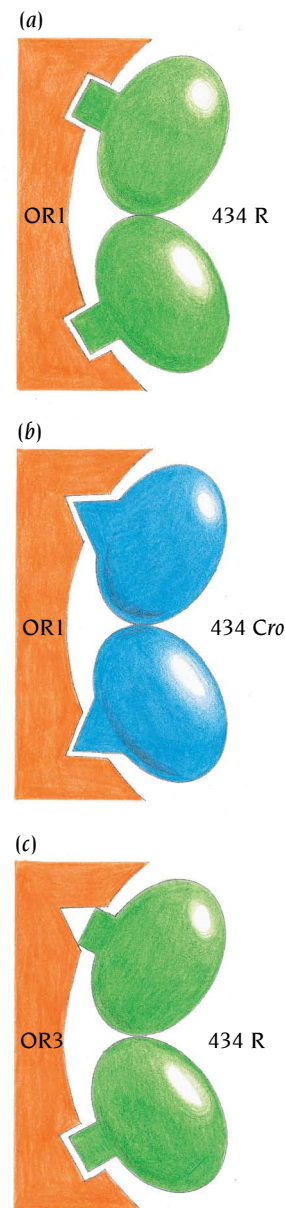


Figure 8.13 (right) Binding of 434 repressor fragment (a) and 434 Cro (b) to operator region OR1. This binding induces different structural changes in the region of the DNA that binds the proteins illustrated as different shapes of the binding regions of OR1. In the complex of operator OR3 with the 434 repressor fragment (c) the two half sites of OR3 are different: one is similar to OR1, with bound repressor, whereas the other has a different nucleotide sequence that adopts the Cro-type binding conformation on binding repressor. The binding surfaces of the DNA and repressor fragment do not complement each other as they do in the OR1 complex; consequently, the repressor fragment binds more weakly to OR3 than to OR1. DNA is schematically shown in orange and the proteins in blue or green.

(Figure 8.14). There are no interactions between the protein and the bases of the DNA in this middle region of the operator.

Residues of the recognition α helix project their side chains into the major groove and interact with the edges of the DNA base pairs on the floor of the groove (Figure 8.15a). Gln (Q) 28 forms two hydrogen bonds to N_6 and N_7 of A1 in base pair 1 (T14'–A1) (Figure 8.15b), and Gln 29 forms a hydrogen bond to O_6 of G13' in base pair 2 (G13'–C2). At base pair 3 (T12'–A3) no hydrogen bonding to the protein occurs and direct contacts are all hydrophobic. The methyl groups of the side chains of Thr (T) 27 and Gln (Q) 29 form a hydrophobic pocket to receive the methyl group of T12'.

The first three base pairs in all six operator regions recognized by phage 434 repressor are identical (see Table 8.2). This means that interactions between these three base pairs and the two glutamine residues (28 and 29) cannot contribute to the discrimination between the six binding sites in the DNA; rather, these interactions provide a general recognition site for operator regions. This simple pattern of hydrogen bonds and hydrophobic interactions therefore accounts for the specificity of phage 434 Cro and repressor proteins for 434 operator regions. The role of Gln 29 is particularly important since it interacts with both base pairs 2 and 3. Its hydrogen bond to guanine specifies C–G at position 2 and the hydrophobic pocket formed by the hydrophobic part of its side chain together with residue 27, specifies A–T at base pair 3.

This general recognition function is crucial for the bacteriophage. When glutamines 28 and 29 are replaced by any other amino acid, the mutant phages are no longer viable. Moreover, 434 Cro protein has glutamine residues at these two positions in its recognition helix as expected since it binds to the same set of operator regions.

Protein–DNA backbone interactions determine DNA conformation

It is apparent from the crystal structures of these protein–DNA complexes that the differential affinities of 434 repressor and Cro for the different operator regions are not determined by sequence-specific interactions between amino acid side chains of the recognition helix and base pairs in the major groove of DNA. Instead, they seem to be determined mainly by the ability of the DNA to undergo specific structural changes so that complementary surfaces are formed between the proteins and the DNA. Interactions between the DNA sugar-phosphate backbone and the proteins are important for establishing such structural changes and thus for positioning the recognition helix correctly in the major groove.

The interface between the repressor fragment and DNA covers an extended area. Most interactions between one monomer of the fragment and the DNA occur within a single half-site; in other words, each repressor subunit

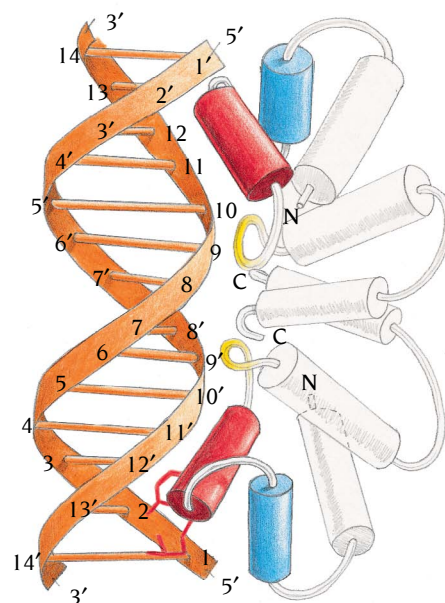


Figure 8.14 Overall view of the complex between 434 repressor fragment and a palindromic synthetic 14mer of DNA (see Table 8.2). The two binding sites of the repressor dimer to the DNA are identical. The recognition helices of the repressor are red, and the first helix of the helix-turn-helix motif is blue. (Adapted from J. Anderson et al., *Nature* 326: 846–852, 1987.)

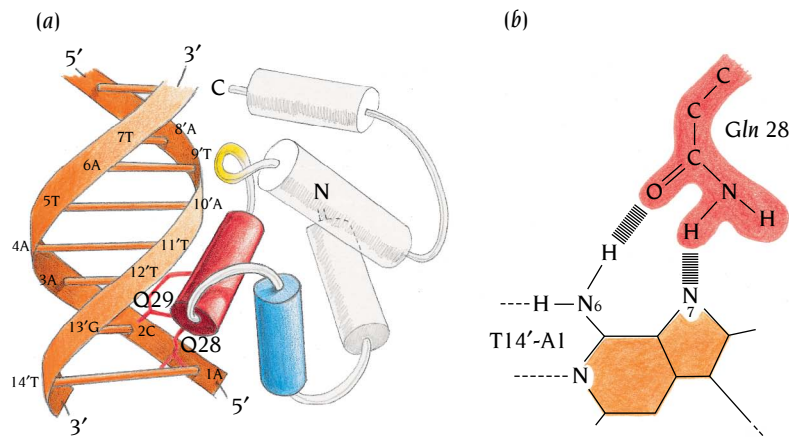


Figure 8.15 Sequence-specific protein–DNA interactions provide a general recognition signal for operator regions in 434 bacteriophage. (a) In this complex between 434 repressor fragment and a synthetic DNA there are two glutamine residues (28 and 29) at the beginning of the recognition helix in the helix-turn-helix motif that provide such interactions with the first three base pairs of the operator region. The side chain of Gln 28 forms two hydrogen bonds (b) to the edge of the adenine base of base pair T14'–A1 in the major groove of the DNA. (Adapted from J. Anderson et al., *Nature* 326: 846–852, 1987.)